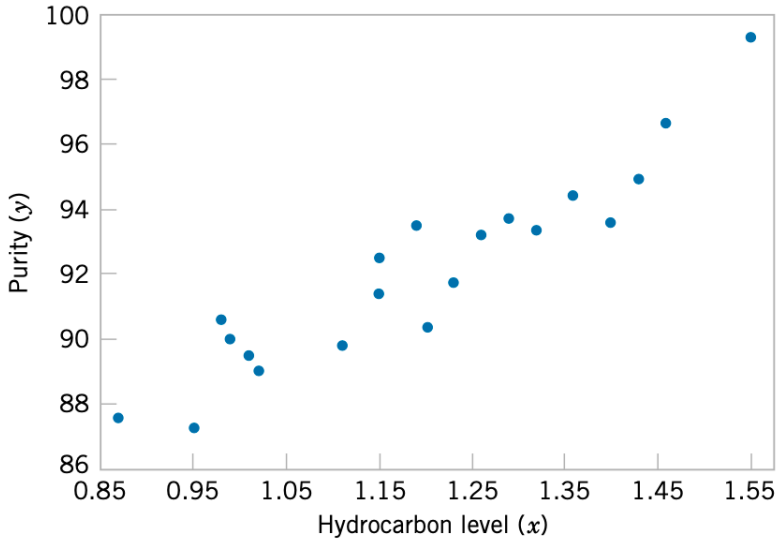
STA 2002, Summer 2024
Probability and Statistics II

Simple Linear Regression

Example: Scatter Diagram

- Many problems in statistics and science involve exploring the relationships between two or more variables.
- **Regression analysis** is a statistical technique that is very useful for these types of problems.





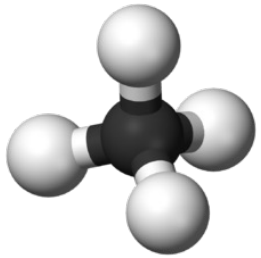


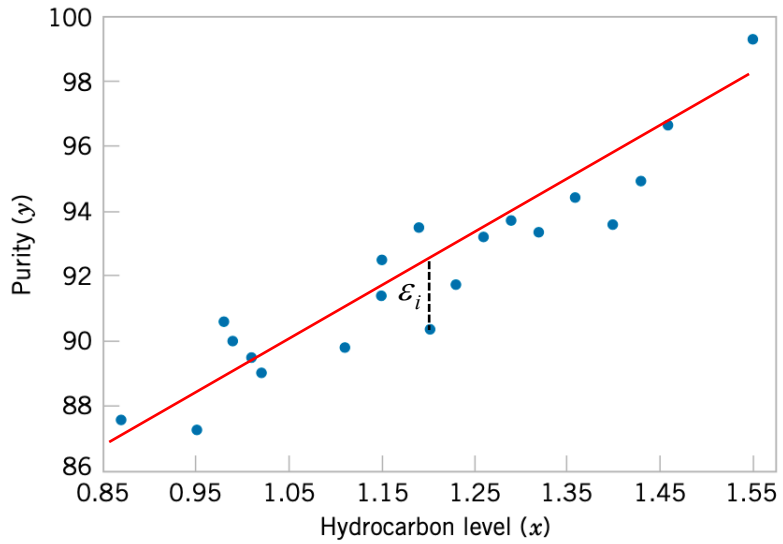
Table 11-1 Oxygen and Hydrocarbon Levels

Observation Number	Hydrocarbon Level x(%)	Purity y(%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
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15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
18	1.32	93.41
19	1.43	94.98
20	0.95	87.33

Reading: Chapter 6.5

Simple Linear Regression

Based on the scatter diagram, it is reasonable to assume that the mean of the random variable Y is related to X by the following simple linear regression model:



Response — Regressor or Predictor

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad i = 1,2,\cdots,n$$

Intercept — Slope — Random error

$$\varepsilon_i \sim N(0,\sigma^2)$$

Mean response $E(Y|x) = \beta_0 + \beta_1 x$

$$V(Y|x) = \sigma^2$$

the slope β_1 and intercept β_0 of the line are called regression coefficients.

- The case of simple linear regression considers a single regressor or predictor X and a dependent or response variable Y.

Simple Linear Regression

The method of least squares is used to estimate the parameters, β_0 and β_1, by minimizing the sum of the squares of the vertical deviations.

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \qquad i = 1,2,\cdots,n$$

sum of the squares of the error

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} (y_i - \beta_0 - \beta_1 x_i)^2$$

Minimize

$$\left.\frac{\partial L}{\partial \beta_0}\right|_{\hat{\beta}_0,\hat{\beta}_1} = -2\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) = 0$$

$$\left.\frac{\partial L}{\partial \beta_1}\right|_{\hat{\beta}_0,\hat{\beta}_1} = -2\sum_{i=1}^{n} (y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i) x_i = 0$$

$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_i = \sum_{i=1}^{n} y_i$$

$$\hat{\beta}_0 \sum_{i=1}^{n} x_i + \hat{\beta}_1 \sum_{i=1}^{n} x_i^2 = \sum_{i=1}^{n} y_i x_i$$

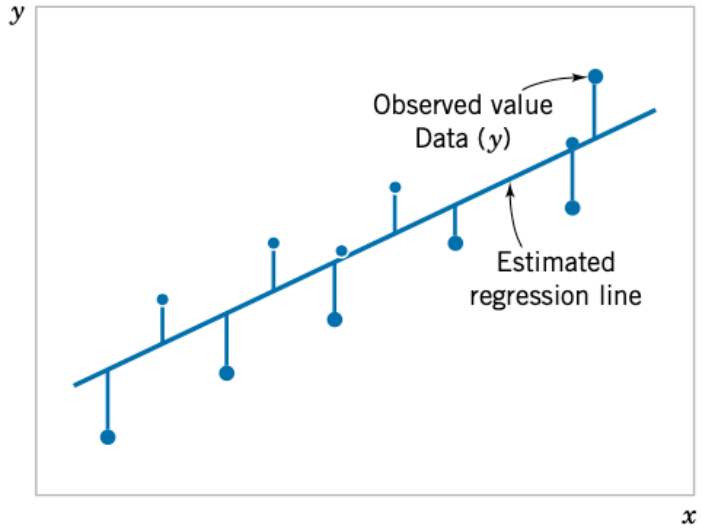


Figure Deviations of the data from the estimated regression model.

Least Square Estimates

The least squares estimates of the intercept and slope in the simple linear regression model are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

Alternative Notation

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} \qquad S_{xy} = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$$\hat{y}_i = \hat{\beta}_0 + \hat{\beta}_1 x_i$$

Fitted (estimated) regression model

$e_i = y_i - \hat{y}_i$ is called the **residual.**

Maximum Likelihood Estimates

$$\left.\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + \varepsilon_i \\ \varepsilon_i &\sim N(0,\sigma^2) \end{aligned}\right\} \quad Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \qquad i = 1,2,\cdots,n$$

Likelihood function $$L(\beta_0,\beta_1) = \left(\frac{1}{\sqrt{2\pi\sigma^2}}\right)^n e^{-\sum(y_i-\beta_0-\beta_1 x_i)^2/(2\sigma^2)}$$

Log-Likelihood function $$\ell(\beta_0,\beta_1) = -nln\left(\sqrt{2\pi\sigma^2}\right) - \sum (y_i - \beta_0 - \beta_1 x_i)^2/(2\sigma^2)$$

Maximum Likelihood Estimate

$$\left.\begin{aligned} \frac{\partial \ell(\beta_0,\beta_1)}{\partial \beta_0}\Big|_{\hat\beta_0,\hat\beta_1} &= \sum \left(y_i - \hat\beta_0 - \hat\beta_1 x_i\right) \ /\sigma^2 = 0 \\ \frac{\partial \ell(\beta_0,\beta_1)}{\partial \beta_1}\Big|_{\hat\beta_0,\hat\beta_1} &= \sum x_i \left(y_i - \hat\beta_0 - \hat\beta_1 x_i\right) \ /\sigma^2 = 0 \end{aligned}\right\}$$

Same as Least Square (LS) Estimate

$$\hat\beta_0 = \bar y - \hat\beta_1 \bar x$$

$$\hat\beta_1 = \frac{\sum_{i=1}^n y_i x_i - \frac{\left(\sum_{i=1}^n y_i\right)\left(\sum_{i=1}^n x_i\right)}{n}}{\sum_{i=1}^n x_i^2 - \frac{\left(\sum_{i=1}^n x_i\right)^2}{n}}$$

Maximum Likelihood Estimates

$$\left.\begin{array}{l} Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i \\ \varepsilon_i \sim N(0,\sigma^2) \end{array}\right\} \quad Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \qquad i = 1,2,\cdots,n$$

Log-Likelihood function $\ell(\beta_0, \beta_1, \sigma^2) = -n ln\left(\sqrt{2\pi\sigma^2}\right) - \sum (y_i - \beta_0 - \beta_1 x_i)^2/(2\sigma^2)$

If $\boldsymbol{\sigma^2}$ **is unknown**, the MLE for σ^2 is:

$$\frac{\partial \ell}{\partial(\sigma^2)}\Big|_{\hat{\beta}_0,\hat{\beta}_1,\hat{\sigma}^2} = -\frac{n}{2}\frac{1}{\hat{\sigma}^2} + \sum \left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2/(2\hat{\sigma}^4) = 0$$

$$\Rightarrow \quad \hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$

We will see that this **MLE is biased**, and we will convert it to an unbiased estimate and use for CI and Hypothesis test.

Example

Find the least square estimates of the simple linear regression describing the relationship between Purity (y) and Hydrocarbon Levels (x).
Also, calculate the predicted purity when hydrocarbon level is 1.00.

$$n = 20 \quad \sum_{i=1}^{20} x_i = 23.92 \quad \sum_{i=1}^{20} y_i = 1{,}843.21 \quad \bar{x} = 1.1960 \quad \bar{y} = 92.1605$$

$$\sum_{i=1}^{20} y_i^2 = 170{,}044.5321 \quad \sum_{i=1}^{20} x_i^2 = 29.2892 \quad \sum_{i=1}^{20} x_i y_i = 2{,}214.6566$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}$$

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17	1.26	93.25
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19	1.43	94.98
20	0.95	87.33

Example: Fitted model

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{10.17744}{0.68088} = 14.94748$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = 92.1605 - (14.94748)1.196 = 74.28331$$

$$\hat{y} = 74.283 + 14.947x$$

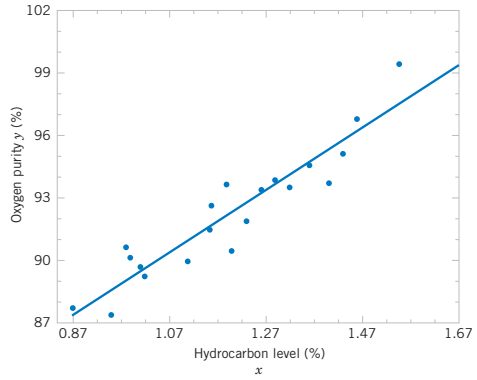
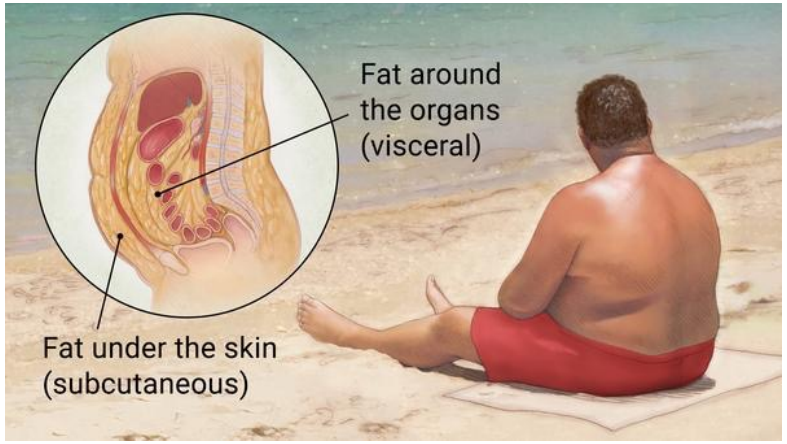


Figure 11-4 Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y} = 74.283 + 14.947x$.

Practical Interpretation: Using the regression model, we would predict oxygen purity of $\hat{y} = 89.23\%$ when the hydrocarbon level is $x = 1.00\%$. The purity 89.23% may be interpreted as an estimate of the true population mean purity when $x = 1.00\%$, or as an estimate of a new observation when $x = 1.00\%$. These estimates are, of course, subject to error; that is, it is unlikely that a future observation on purity would be exactly 89.23% when the hydrocarbon level is 1.00%. In subsequent sections we will see how to use confidence intervals and prediction intervals to describe the error in estimation from a regression model.

Example: Diabetes and Obesity

- Diabetes and obesity are serious health concerns in the US. Measuring the amount of body fat of a person is one way to monitor body weight control.
- BMI (body mass index) = mass (kg) / (height(m))^2 is used as a proxy to body fat.
- In a study of 250 men at Brigham Young U, both BMI x and body fat y were measured. The summary statistics are:



$$\sum_{i=1}^{n} x_i = 6322.28 \qquad \sum_{i=1}^{n} x_i^2 = 162674.18$$

$$\sum_{i=1}^{n} y_i = 4757.90 \qquad \sum_{i=1}^{n} y_i^2 = 107679.27$$

$$\sum_{i=1}^{n} x_i\, y_i = 125471.10$$

Fit a linear regression model

Example: Diabetes and Obesity

Body fat example:

$n = 250$ $\quad \bar{x} = \frac{\sum x_i}{250} = 25.28912$

$\bar{y} = \frac{\sum y_i}{250} = 19.0316$

$S_{xx} = \sum x_i^2 - n\bar{x}^2 = 2789.282$

$S_{xy} = \sum x_i y_i - n\bar{x}\bar{y} = 5147.996$

$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = 1.846 \quad \hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} = -27.652$

Continue: body fat vs BMI

- Use the equation of the fitted line to predict what body fat would be observed, on average, for a man with BMI = 30

For BMI = 30

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x = -27.652 + 1.846 \times 30 = 27.728$$

- Suppose the observed body fat of a man with a BMI of 25 is 25%, find the residual for that observation.

For BMI = 25

$$\hat{y} = \hat{\beta}_0 + \hat{\beta}_1 x = -27.652 + 1.846 \times 25 = 18.498$$

observed $y = 25$

$$\text{residual} = y - \hat{y} = 25 - 18.498 = 6.502$$

Model Diagnosis

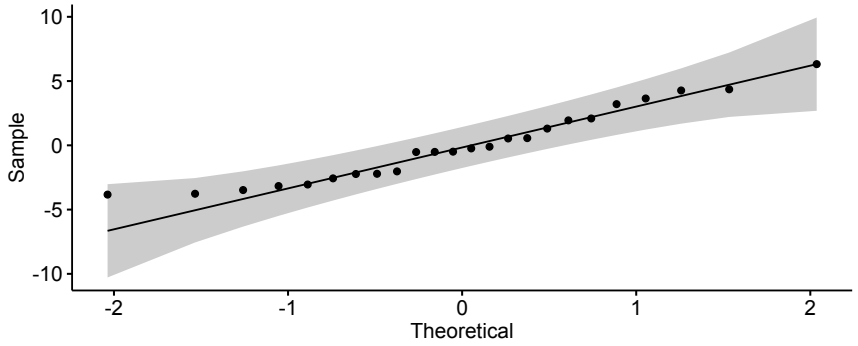
Analysis of Residual Patterns is useful for checking:

- Independency assumption
- Constant variance

Plot residuals (e_i) against predicted response ($\hat{y}_i$)

Histogram for residuals or Q-Q plot are useful for checking:

- Normality assumption



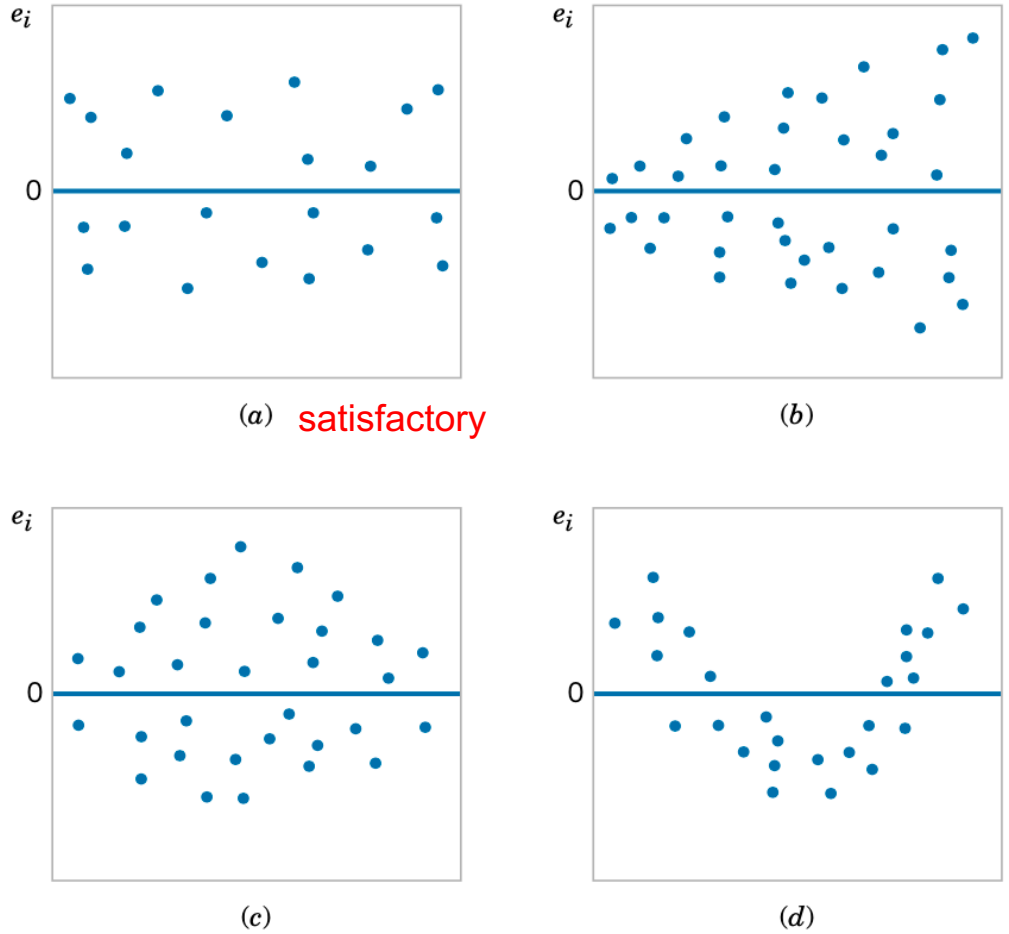


Figure Residual plots. (a) satisfactory, (b) funnel, (c) double bow, (d) nonlinear. [Adapted from Montgomery, Peck, and Vining (2001).]

Confidence Interval

Reading: Chapter 7.6

Maximum likelihood (or LS) estimates

The maximum likelihood (or equivalently, the least squares) estimates of the intercept and slope in the simple linear regression model are

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$$\left.\begin{aligned} Y_i &= \beta_0 + \beta_1 X_i + \varepsilon_i \\ \varepsilon_i &\sim N(0, \sigma^2) \end{aligned}\right\} \quad Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \qquad i = 1, 2, \cdots, n$$

⇒ What is the distribution of $\hat{\beta}_0, \hat{\beta}_1$?

Distribution of the estimates

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x}$$

$$\hat{\beta}_1 = \frac{\sum_{i=1}^{n} y_i x_i - \frac{\left(\sum_{i=1}^{n} y_i\right)\left(\sum_{i=1}^{n} x_i\right)}{n}}{\sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}}$$

where $\bar{y} = (1/n)\sum_{i=1}^{n} y_i$ and $\bar{x} = (1/n)\sum_{i=1}^{n} x_i$.

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}$$

$$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

$$\varepsilon_i \sim N(0, \sigma^2)$$

$$Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \qquad i = 1, 2, \cdots, n$$

$$\hat{\beta}_1 = \frac{1}{S_{xx}} \sum_{i=1}^{n} y_i (x_i - \bar{x})$$

$$\hat{\beta}_0 = \sum_{i=1}^{n} y_i \left(\frac{1}{n} - \frac{\bar{x}}{S_{xx}} (x_i - \bar{x})\right)$$

Distribution of the estimates

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$
$$\varepsilon_i \sim N(0, \sigma^2)$$

$$Y_i \sim N(\beta_0 + \beta_1 X_i, \sigma^2) \qquad i = 1, 2, \cdots, n$$

$$\hat{\beta}_1 = \frac{1}{S_{xx}} \sum_{i=1}^{n} y_i (x_i - \bar{x})$$

$$\hat{\beta}_0 = \sum_{i=1}^{n} y_i \left(\frac{1}{n} - \frac{\bar{x}}{S_{xx}} (x_i - \bar{x}) \right)$$

$$E(\hat{\beta}_1) = \beta_1 \qquad V(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$$

$$E(\hat{\beta}_0) = \beta_0 \quad \text{and} \quad V(\hat{\beta}_0) = \sigma^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right]$$

$$\text{cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2 \bar{x} / S_{xx}.$$

Recall:

- If $x_1 \sim N(\mu_1, \sigma_1^2), x_2 \sim N(\mu_2, \sigma_2^2)$ are independently normally distributed variables, then

$$x_1 + x_2 \sim N(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2 \left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

Estimation of Variance (σ^2)

The maximum likelihood estimates of the unknown variance is biased

$$\hat{\sigma}^2_{\text{MLE}} = \frac{1}{n}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2 \quad \text{MLE (biased)}$$

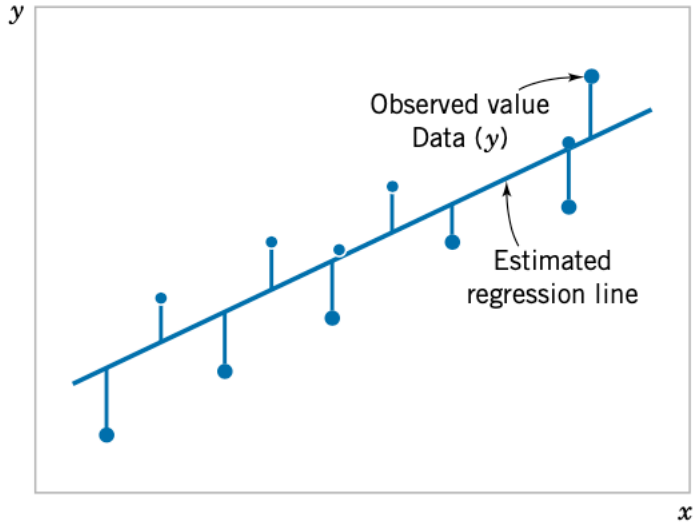
The **error sum of squares** is

$$SS_E = \sum_{i=1}^{n} e_i^2 = \sum_{i=1}^{n}\left(y_i - \hat{y}_i\right)^2$$

$$SS_E = \sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$

$$\frac{\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2}{\sigma^2} \sim \chi^2(n-2)$$

$$\Rightarrow \quad E(SS_E) = (n-2)\sigma^2.$$



An unbiased estimator of σ^2 is

$$\hat{\sigma}^2 = \frac{SS_E}{n-2}$$

where SS_E can be easily computed using (easier formula)

$$SS_E = SS_T - \hat{\beta}_1 S_{xy}$$

$$SS_T = \sum_{i=1}^{n}(y_i - \bar{y})^2 = \sum_{i=1}^{n} y_i^2 - \frac{\left(\sum_{i=1}^{n} y_i\right)^2}{n}$$

Total sum of square for y

Example: estimate variance

$$n = 20 \quad \sum_{i=1}^{20} x_i = 23.92 \quad \sum_{i=1}^{20} y_i = 1{,}843.21 \quad \bar{x} = 1.1960 \quad \bar{y} = 92.1605$$

$$\sum_{i=1}^{20} y_i^2 = 170{,}044.5321 \quad \sum_{i=1}^{20} x_i^2 = 29.2892 \quad \sum_{i=1}^{20} x_i y_i = 2{,}214.6566$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n} \qquad \hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} = \frac{10.17744}{0.68088} = 14.94748$$

$$S_{xy} = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}$$

Example on variance estimation:

oxygen example:

$$SS_T = \Sigma (y_i - \bar{y})^2 = \Sigma y_i^2 - n\bar{y}^2 = 173.3769$$

$$SS_E = SS_T - \hat{\beta}_1 S_{xy} = 21.2498$$

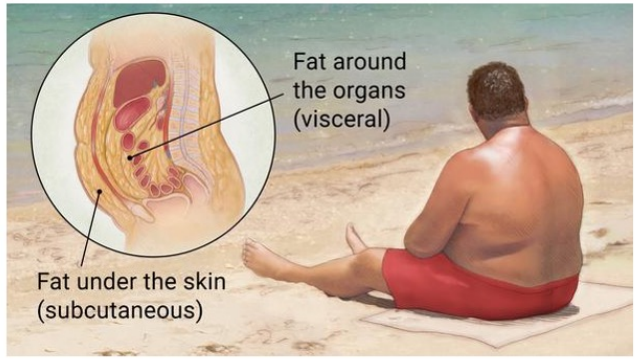
$$\hat{\sigma}^2 = \frac{21.2498}{n-2} = 1.18$$

Table 11-1 Oxygen and Hydrocarbon Levels

Observation Number	Hydrocarbon Level x (%)	Purity y (%)
1	0.99	90.01
2	1.02	89.05
3	1.15	91.43
4	1.29	93.74
5	1.46	96.73
6	1.36	94.45
7	0.87	87.59
8	1.23	91.77
9	1.55	99.42
10	1.40	93.65
11	1.19	93.54
12	1.15	92.52
13	0.98	90.56
14	1.01	89.54
15	1.11	89.85
16	1.20	90.39
17	1.26	93.25
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19	1.43	94.98
20	0.95	87.33

Example: estimate variance

- In a study of 250 men at Brigham Young U, both BMI x and body fat y were measured. The summary statistics are:



$$\sum_{i=1}^{n} x_i = 6322.28 \qquad \sum_{i=1}^{n} x_i^2 = 162674.18$$

$$\sum_{i=1}^{n} y_i = 4757.90 \qquad \sum_{i=1}^{n} y_i^2 = 107679.27$$

$$\sum_{i=1}^{n} x_i\, y_i = 125471.10$$

Fit a linear regression model

Body fat example:

$$SS_T = \Sigma y_i^2 - n\bar{y}^2 = 17128.82$$

$$SS_E = SS_T - \hat{\beta}_1 S_{xy} = 7625.62$$

$$\hat{\sigma}^2 = \frac{SSE}{n-2} = \frac{SSE}{248} = 30.75$$

Distribution of the estimate

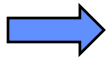
$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$$

$S_R^2, \hat{\beta}_0, \hat{\beta}_1$ are mutually independent

$$\hat{\sigma}^2 := \frac{1}{n-2}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$$

$$\frac{\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2}{\sigma^2} \sim \chi^2(n-2)$$



$$\frac{\hat{\beta}_1 - \beta_1}{\hat{\sigma}/\sqrt{S_{xx}}} \sim t(n-2)$$

$$\frac{\hat{\beta}_0 - \beta_0}{\hat{\sigma}/\sqrt{\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}}} \sim t(n-2)$$

Confidence Interval for Regression Coefficients

Mean and variance of the slope estimator $E(\hat{\beta}_1) = \beta_1$ $V(\hat{\beta}_1) = \frac{\sigma^2}{S_{xx}}$

Use this to find standard error

Under the assumption that the observations are normally and independently distributed, a $100(1-\alpha)\%$ **confidence interval on the slope** β_1 in simple linear regression is

$$\hat{\beta}_1 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \leq \beta_1 \leq \hat{\beta}_1 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

Mean and variance of the intercept estimator $E(\hat{\beta}_0) = \beta_0$ and $V(\hat{\beta}_0) = \sigma^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]$

Similarly, a $100(1-\alpha)\%$ **confidence interval on the intercept** β_0 is

$$\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}\right)} \leq \beta_0 \leq \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{\bar{x}^2}{S_{xx}}\right)}$$

$\text{cov}(\hat{\beta}_0, \hat{\beta}_1) = -\sigma^2\bar{x}/S_{xx}.$

The width of confidence interval indicates the overall quality of regression line.

Gas purity example

EXAMPLE 11-4 Oxygen Purity Confidence Interval on the Slope

We will find a 95% confidence interval on the slope of the regression line using the data in Example 11-1. Recall that $\hat{\beta}_1 = 14.947$, $S_{xx} = 0.68088$, and $\hat{\sigma}^2 = 1.18$ (see Table 11-2). Then, from Equation 11-29 we find

$$\hat{\beta}_1 - t_{0.025,18}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}} \le \beta_1 \le \hat{\beta}_1 + t_{0.025,18}\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}$$

or

$$14.947 - 2.101\sqrt{\frac{1.18}{0.68088}} \le \beta_1 \le 14.947 + 2.101\sqrt{\frac{1.18}{0.68088}}$$

This simplifies to

$$12.181 \le \beta_1 \le 17.713$$

Practical Interpretation: This CI does not include zero, so there is strong evidence (at $\alpha = 0.05$) that the slope is not zero. The CI is reasonably narrow (± 2.766) because the error variance is fairly small.



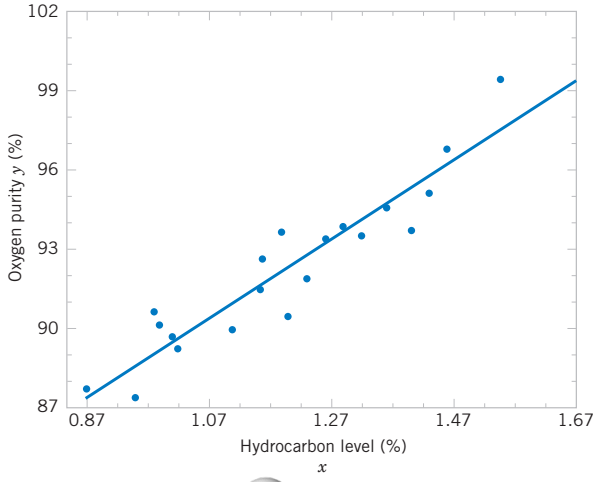
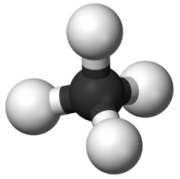


Figure 11-4 Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y} = 74.283 + 14.947x$.



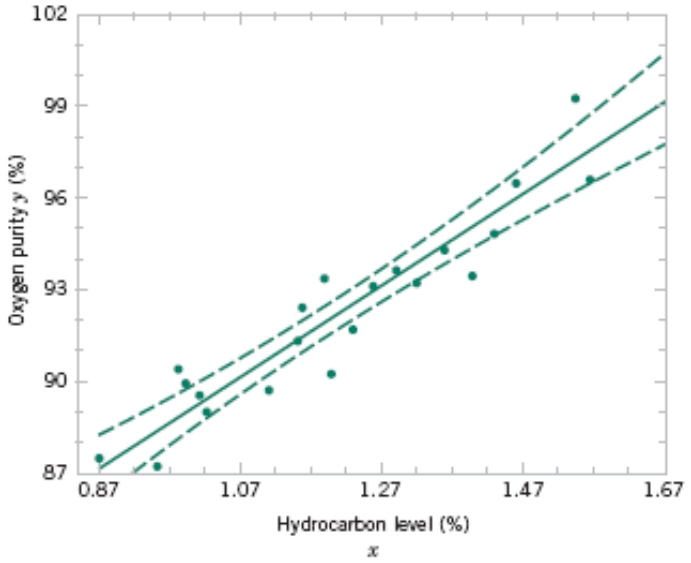
Confidence Interval on Mean Response

A $100(1-\alpha)\%$ **confidence interval about the mean response** at the value of $x = x_0$, say $\mu_{Y|x_0}$, is given by

$$\hat{\mu}_{Y|x_0} - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$

$$\leq \mu_{Y|x_0} \leq \hat{\mu}_{Y|x_0} + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$

where $\hat{\mu}_{Y|x_0} = \hat{\beta}_0 + \hat{\beta}_1 x_0$ is computed from the fitted regression model.



$$E(Y|x_0) = \mu_{Y|x_0} = \beta_0 + \beta_1 x_0$$

EXAMPLE 11-5 Oxygen Purity Confidence Interval on the Mean Response

We will construct a 95% confidence interval about the mean response for the data in Example 11-1. The fitted model is $\hat{\mu}_{Y|x_0} = 74.283 + 14.947x_0$, and the 95% confidence interval on $\mu_{Y|x_0}$ is found from Equation 11-31 as

$$\hat{\mu}_{Y|x_0} \pm 2.101\sqrt{1.18\left[\frac{1}{20} + \frac{(x_0 - 1.1960)^2}{0.68088}\right]}$$

Suppose that we are interested in predicting mean oxygen purity when $x_0 = 1.00\%$. Then

$$\hat{\mu}_{Y|x_{1.00}} = 74.283 + 14.947(1.00) = 89.23$$

and the 95% confidence interval is

$$89.23 \pm 2.101\sqrt{1.18\left[\frac{1}{20} + \frac{(1.00 - 1.1960)^2}{0.68088}\right]}$$

or

$$89.23 \pm 0.75$$

Therefore, the 95% CI on $\mu_{Y|1.00}$ is

$$88.48 \leq \mu_{Y|1.00} \leq 89.98$$



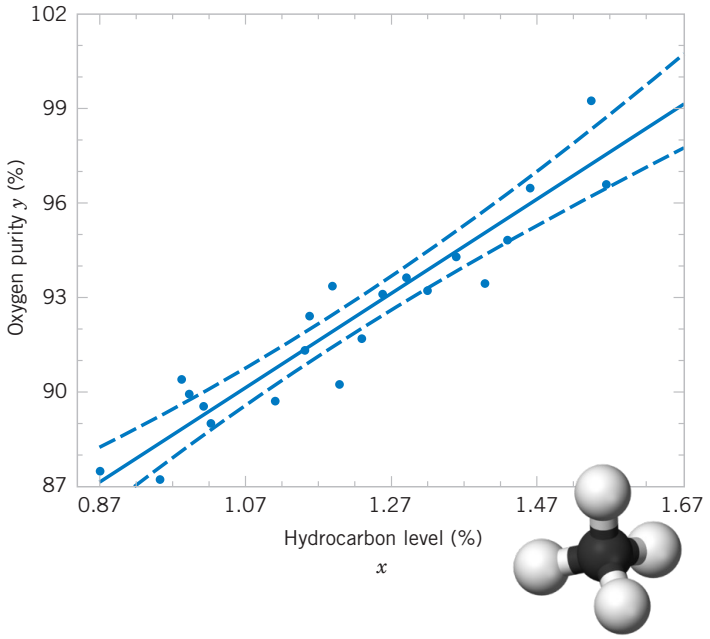


Figure 11-7 Scatter diagram of oxygen purity data from Example 11-1 with fitted regression line and 95 percent confidence limits on $\mu_{Y|x_0}$.

Prediction interval

- Predicting response for **new** observation
- The **new** observation is **independent** of data used to build linear regression model

A $100(1-\alpha)$ % **prediction interval on a future observation** Y_0 at the value x_0 is given by

$$\hat{y}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)} \leq Y_0 \leq \hat{y}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1+\frac{1}{n}+\frac{(x_0-\bar{x})^2}{S_{xx}}\right)}$$

The value $\hat{y}_0$ is computed from the regression model $\hat{y}_0 = \hat{\beta}_0 + \hat{\beta}_1 x_0$.

Gas Purity Example

EXAMPLE 11-6 **Oxygen Purity Prediction Interval**

To illustrate the construction of a prediction interval, suppose we use the data in Example 11-1 and find a 95% prediction interval on the next observation of oxygen purity at $x_0 = 1.00\%$. Using Equation 11-33 and recalling from Example 11-5 that $\hat{y}_0 = 89.23$, we find that the prediction interval is

$$89.23 - 2.101\sqrt{1.18\left[1 + \frac{1}{20} + \frac{(1.00 - 1.1960)^2}{0.68088}\right]}$$

$$\leq Y_0 \leq 89.23 + 2.101\sqrt{1.18\left[1 + \frac{1}{20} + \frac{(1.00 - 1.1960)^2}{0.68088}\right]}$$

which simplifies to

$$86.83 \leq y_0 \leq 91.63$$

This is a reasonably narrow prediction interval.

Minitab will also calculate prediction intervals. Refer to the output in Table 11-2. The 95% PI on the future observation at $x_0 = 1.00$ is shown in the display.

By repeating the foregoing calculations at different levels of x_0, we may obtain the 95% prediction intervals shown graphically as the lower and upper lines about the fitted regression model in Fig. 11-8. Notice that this graph also shows the 95% confidence limits on $\mu_{Y|x_0}$ calculated in Example 11-5. It illustrates that the prediction limits are always wider than the confidence limits.

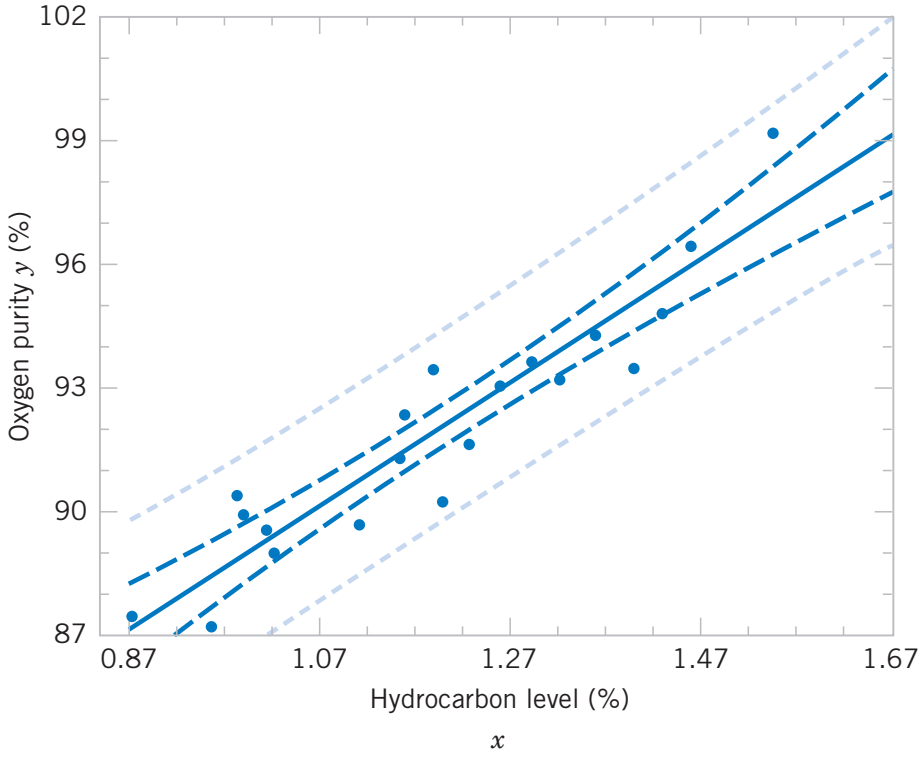


Figure 11-8 Scatter diagram of oxygen purity data from Example 11-1 with fitted regression line, 95% prediction limits (outer lines) and 95% confidence limits on $\mu_{Y|x_0}$.

Observation:

- Prediction interval is always **wider** than confidence interval. This is because prediction interval depends on both the **error from fitting the model** and the **randomness of the new observation**.

An Alternative Perspective

- Sometimes, in order to simplify the model and computation, we may write

New Regressor or Predictor X_i'

$$Y_i = (\beta_0 + \beta_1 \bar{X}) + \beta_1 (X_i - \bar{X}) + \epsilon_i$$

New Intercept β_0' — Slope (the same) — Random error

$$\varepsilon_i \sim N(0, \sigma^2)$$

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Original model

- So the model now becomes:

$$Y_i = \beta_0 + \beta_1 (X_i - \bar{X}) + \epsilon_i$$

- The least-square estimate, maximum likelihood estimate, distributions of estimate, confidence intervals, can be obtained by directly applying previous results (with new intercept and new regressor variables X').

An Alternative Perspective

- Under the alternative model

$$Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i$$

- The least-square (also the maximum likelihood) estimate:

$$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}}$$

$$\hat{\beta}_0 = \bar{y}$$

$$S_{xx} = \sum_{i=1}^{n} (x_i - \bar{x})^2 = \sum_{i=1}^{n} x_i^2 - \frac{\left(\sum_{i=1}^{n} x_i\right)^2}{n}$$

$$S_{xy} = \sum_{i=1}^{n} (y_i - \bar{y})(x_i - \bar{x}) = \sum_{i=1}^{n} x_i y_i - \frac{\left(\sum_{i=1}^{n} x_i\right)\left(\sum_{i=1}^{n} y_i\right)}{n}$$

- Distributions:

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$$

$$\hat{\beta}_0 \sim N\left(\beta_0, \frac{\sigma^2}{n}\right)$$

An Alternative Perspective

- Distribution of the variance estimate

$$Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i$$

error sum of squares

Note:

$$\frac{\sum_{i=1}^{n}(y_i - \beta_0 - \beta_1(x_i - \bar{x}))^2}{\sigma^2} = \frac{\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1(x_i - \bar{x})\right)^2}{\sigma^2} + \frac{n}{\sigma^2}\left(\hat{\beta}_0 - \beta_0\right)^2 + \frac{S_{xx}}{\sigma^2}\left(\hat{\beta}_1 - \beta_1\right)^2$$

$$\sim \chi^2(n) \qquad \sim \chi^2(1) \qquad \sim \chi^2(1)$$

$$\hat{\beta}_1 \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right) \qquad \hat{\beta}_0 \sim N\left(\beta_0, \frac{\sigma^2}{n}\right)$$

$$\Rightarrow \quad \frac{\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1(x_i - \bar{x})\right)^2}{\sigma^2} \sim \chi^2(n-2)$$

$$\Rightarrow \quad \hat{\sigma}^2 := \frac{1}{n-2}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1(x_i - \bar{x})\right)^2$$

Unbiased estimate for σ^2

An Alternative Perspective

- Confidence Interval

$$Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i$$

Intercept $$[\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}, \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}]$$

Slope $$[\hat{\beta}_1 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \hat{\beta}_1 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}]$$

- Confidence Interval on mean response

$$[\hat{\mu}_{Y|x_0} - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}, \hat{\mu}_{Y|x_0} + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}]$$

$$\hat{y}_0 = \hat{\mu}_{Y|x_0} = \hat{\beta}_0 + \hat{\beta}_1(x_0 - \bar{x})$$

- Prediction Interval

$$[\hat{y}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}, \hat{y}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(1 + \frac{1}{n} + \frac{(x_0 - \bar{x})^2}{S_{xx}}\right)}]$$

Summary

Regression model $Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$ $\quad \varepsilon_i \sim N(0,\sigma^2)$

Parameter	Estimate	Confidence interval
β_0	$\hat{\beta}_0 = \bar{y} - \hat{\beta}_1 \bar{x} \sim N\left(\beta_0, \sigma^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)\right)$	$[\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}, \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}]$
β_1	$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$	$[\hat{\beta}_1 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \hat{\beta}_1 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}]$
σ^2	$\hat{\sigma}^2 := \frac{1}{n-2}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1 x_i\right)^2$	

$$\frac{\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1(x_i - \bar{x})\right)^2}{\sigma^2} \sim \chi^2(n-2)$$

Summary

Regression model $Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i$ $\quad \varepsilon_i \sim N(0,\sigma^2)$

Parameter	Estimate	Confidence interval
β_0	$\hat{\beta}_0 = \bar{y} \sim N\left(\beta_0, \frac{\sigma^2}{n}\right)$	$[\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}, \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}]$
β_1	$\hat{\beta}_1 = \frac{S_{xy}}{S_{xx}} \sim N\left(\beta_1, \frac{\sigma^2}{S_{xx}}\right)$	$[\hat{\beta}_1 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \hat{\beta}_1 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}]$
σ^2	$\hat{\sigma}^2 := \frac{1}{n-2}\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1(x_i - \bar{x})\right)^2$	

$$\frac{\sum_{i=1}^{n}\left(y_i - \hat{\beta}_0 - \hat{\beta}_1(x_i - \bar{x})\right)^2}{\sigma^2} \sim \chi^2(n-2)$$

Hypothesis Test

Reading: Chapter 9.6

Hypothesis Test on Regression Parameters

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Original model

Slope:

Suppose we wish to test

$$H_0: \beta_1 = \beta_{1,0}$$
$$H_1: \beta_1 \neq \beta_{1,0}$$

An appropriate test statistic would be

$$t_0 = \frac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\hat{\sigma}^2 / S_{xx}}}$$

Intercept:

Suppose we wish to test

$$H_0: \beta_0 = \beta_{0,0}$$
$$H_1: \beta_0 \neq \beta_{0,0}$$

An appropriate test statistic would be

$$t_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{\sqrt{\hat{\sigma}^2 \left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}} \right]}}$$

We would **reject the null hypothesis if**

$$|t_0| > t_{\alpha/2, n-2}$$

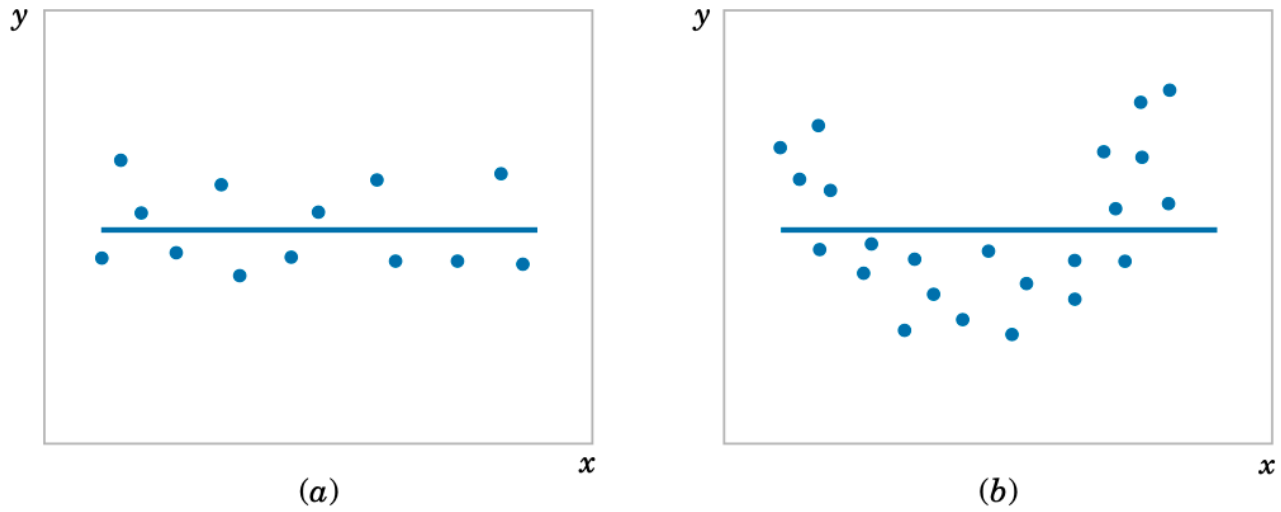
Confidence Intervals can also be used to test the above hypotheses.

Hypothesis Test on Slope

An important special case of the hypotheses on the slope is

$$H_0: \beta_1 = 0$$
$$H_1: \beta_1 \neq 0$$

These hypotheses relate to the significance of regression.
Failure to reject H_0 is equivalent to concluding that there is no linear relationship between *x* and *Y*.



Gas purity example

EXAMPLE 11-2 **Oxygen Purity Tests of Coefficients**

We will test for significance of regression using the model for the oxygen purity data from Example 11-1. The hypotheses are

$$H_0: \beta_1 = 0$$
$$H_1: \beta_1 \neq 0$$

and we will use $\alpha = 0.01$. From Example 11-1 and Table 11-2 we have

$$\hat{\beta}_1 = 14.947 \quad n = 20, \quad S_{xx} = 0.68088, \quad \hat{\sigma}^2 = 1.18$$

so the t-statistic in Equation 10-20 becomes

$$t_0 = \frac{\hat{\beta}_1}{\sqrt{\hat{\sigma}^2/S_{xx}}} = \frac{\hat{\beta}_1}{se(\hat{\beta}_1)} = \frac{14.947}{\sqrt{1.18/0.68088}} = 11.35$$

Practical Interpretation: Since the reference value of t is $t_{0.005,18} = 2.88$, the value of the test statistic is very far into the critical region, implying that H_0: $\beta_1 = 0$ should be rejected. There is strong evidence to support this claim. The P-value for this test is $P \simeq 1.23 \times 10^{-9}$. This was obtained manually with a calculator.



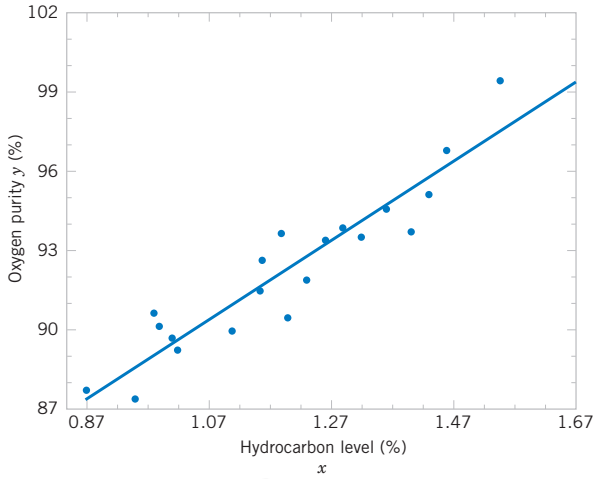
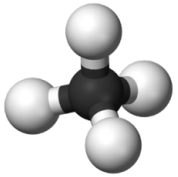


Figure 11-4 Scatter plot of oxygen purity y versus hydrocarbon level x and regression model $\hat{y} = 74.283 + 14.947x$.



Summary: Inference on the Slope

Null Hypothesis $H_0: \beta_1 = \beta_{1,0}$

Test Statistic $t_0 = \dfrac{\hat{\beta}_1 - \beta_{1,0}}{\sqrt{\hat{\sigma}^2/S_{xx}}}$

Distribution under H_0 $t_0 \sim t(n-2)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \beta_1 \neq \beta_{1,0}$	$\lvert t_0 \rvert > t_{\alpha/2}(n-2)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\hat{\beta}_1 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, \hat{\beta}_1 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}]$
$H_1: \beta_1 > \beta_{1,0}$	$t_0 > t_\alpha(n-2)$	$\mathrm{P}(T > t_0)$	$[\hat{\beta}_1 - t_\alpha(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}, +\infty)$
$H_1: \beta_1 < \beta_{1,0}$	$t_0 < -t_\alpha(n-2)$	$\mathrm{P}(T < t_0)$	$(-\infty, \hat{\beta}_1 + t_\alpha(n-2)\sqrt{\frac{\hat{\sigma}^2}{S_{xx}}}]$

Threshold: Critical values

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include $\beta_{1,0}$

Summary: Inference on the Intercept

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Original model

Null Hypothesis

$$H_0: \beta_0 = \beta_{0,0}$$

Test Statistic

$$t_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{\sqrt{\hat{\sigma}^2\left[\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right]}}$$

Distribution under H_0

$$t_0 \sim t(n-2)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
Distribution under H_0	$\lvert t_0\rvert > t_{\alpha/2}(n-2)$	$2\mathrm{P}(T > \lvert t_0\rvert)$	$[\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}, \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}]$
$H_1: \beta_0 > \beta_{0,0}$	$t_0 > t_\alpha(n-2)$	$\mathrm{P}(T > t_0)$	$[\hat{\beta}_0 - t_\alpha(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}, +\infty)$
$H_1: \beta_0 < \beta_{0,0}$	$t_0 < -t_\alpha(n-2)$	$\mathrm{P}(T < t_0)$	$(-\infty, \hat{\beta}_0 + t_\alpha(n-2)\sqrt{\hat{\sigma}^2\left(\frac{1}{n} + \frac{\bar{x}^2}{S_{xx}}\right)}]$

Exercise: what is the test statistic and rejection region under the alternative model?

$$Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i$$

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include $\beta_{0,0}$

Summary: Inference on the Intercept

$$Y_i = \beta_0 + \beta_1(X_i - \bar{X}) + \epsilon_i$$

Null Hypothesis

$H_0: \beta_0 = \beta_{0,0}$

Test Statistic

$t_0 = \frac{\hat{\beta}_0 - \beta_{0,0}}{\hat{\sigma}/\sqrt{n}}$

Distribution under H_0

$t_0 \sim t(n-2)$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value	Confidence interval Approach
$H_1: \beta_0 \neq \beta_{0,0}$	$\lvert t_0 \rvert > t_{\alpha/2}(n-2)$	$2\mathrm{P}(T > \lvert t_0 \rvert)$	$[\hat{\beta}_0 - t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}, \hat{\beta}_0 + t_{\alpha/2}(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}]$
$H_1: \beta_0 > \beta_{0,0}$	$t_0 > t_\alpha(n-2)$	$\mathrm{P}(T > t_0)$	$[\hat{\beta}_0 - t_\alpha(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}, +\infty)$
$H_1: \beta_0 < \beta_{0,0}$	$t_0 < -t_\alpha(n-2)$	$\mathrm{P}(T < t_0)$	$(-\infty, \hat{\beta}_0 + t_\alpha(n-2)\sqrt{\frac{\hat{\sigma}^2}{n}}]$

H_0 is rejected when p-value<α

H_0 is rejected when the CI does NOT include $\beta_{0,0}$

Testing Correlation

Reading: Chapter 9.6

Recap: Covariance and correlation

- Let X, Y be two random variables with mean μ_X, μ_Y, and variance σ_X^2, σ_Y^2
- The **covariance** between X and Y is

$$\text{Cov}(X, Y) = E[(X - \mu_X)(Y - \mu_Y)] = \sigma_{XY}$$

- The **correlation coefficient** between X and Y is

$$\rho = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y} = \frac{\sigma_{XY}}{\sigma_X \sigma_Y}$$

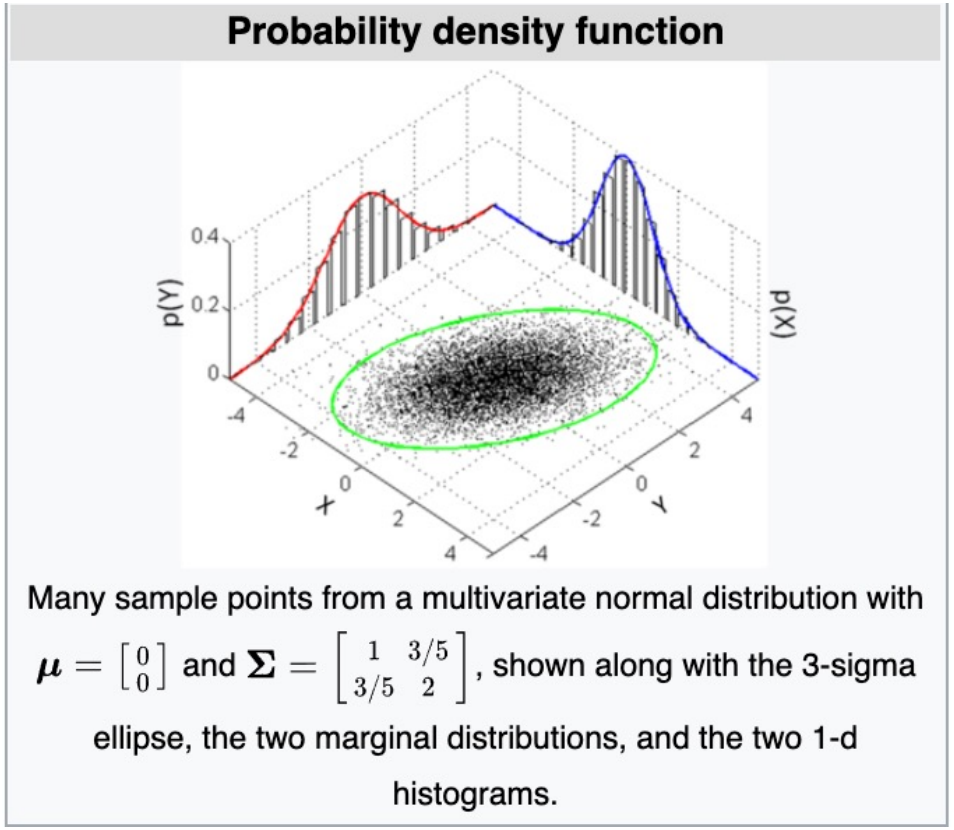
- Fact: if X and Y are independent, then $\sigma_{XY} = \rho = 0$

$$H_0: \rho = 0$$

For normal random variables, the inverse is also true: i.e., if $\rho = 0$ for two normal random variables, then they are independent.

Recap: Bivariate normal

- Let (X, Y) have a bivariate normal distribution
- The marginal distribution of X and Y are also normal distribution
- Let the mean of X and Y be μ_X, μ_Y, and their variances be σ_X^2, σ_Y^2
- The correlation between X and Y is ρ



Many sample points from a multivariate normal distribution with $\boldsymbol{\mu} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}$ and $\boldsymbol{\Sigma} = \begin{bmatrix} 1 & 3/5 \\ 3/5 & 2 \end{bmatrix}$, shown along with the 3-sigma ellipse, the two marginal distributions, and the two 1-d histograms.

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right)$$

Recap: Bivariate normal

$$f(x,y) = \frac{1}{2\pi\sigma_X\sigma_Y\sqrt{1-\rho^2}} \exp\left(-\frac{1}{2(1-\rho^2)}\left[\left(\frac{x-\mu_X}{\sigma_X}\right)^2 - 2\rho\left(\frac{x-\mu_X}{\sigma_X}\right)\left(\frac{y-\mu_Y}{\sigma_Y}\right) + \left(\frac{y-\mu_Y}{\sigma_Y}\right)^2\right]\right)$$

- Let (X, Y) have a bivariate normal distribution
- The conditional distribution of Y given X=x is also a normal distribution

$$Y|X = x \sim N\left(\mu_Y + \rho\frac{\sigma_Y}{\sigma_X}(x-\mu_X), (1-\rho^2)\sigma_Y^2\right)$$

Testing correlation

- Let X, Y have a bivariate normal distribution
- Observe *n* pairs of data (each pair is sampled from the bivariate normal distribution and independent from other pairs):

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$$

- Test:

$$H_0: \rho = 0 \quad \text{v.s.} \quad H_1: \rho \neq 0 \quad H_1: \rho > 0 \quad H_1: \rho < 0$$

Null hypothesis

Testing correlation

- Let X, Y have a bivariate normal distribution
- Rewrite them as:

$$(X_1, Y_1), (X_2, Y_2), \ldots, (X_n, Y_n)$$

$$Y_i = \beta_0 + \beta_1 X_i + \varepsilon_i$$

Null hypothesis

$$\rho = 0 \Rightarrow 0 = Cov(Y_i, X_i) = Cov(\beta_0 + \beta_1 X_i + \epsilon_i, X_i)$$

$$= Cov(\beta_0, X_i) + \beta_1 Cov(X_i, X_i) + Cov(\epsilon_i, X_i)$$

$$= 0 + \beta_1 Var(X_i) + 0$$

$$\Rightarrow \beta_1 = 0$$

Sample Correlation Coefficient

Define sample correlation coefficient:

$$R = \frac{[1/(n-1)]\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{[1/(n-1)]\sum_{i=1}^{n}(X_i - \overline{X})^2}\sqrt{[1/(n-1)]\sum_{i=1}^{n}(Y_i - \overline{Y})^2}} = \frac{S_{XY}}{S_X S_Y}.$$

Sample covariance:

$$s_{XY} = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})(Y_i - \bar{Y})$$

Sample variances:

$$s_X^2 = \frac{1}{n-1}\sum_{i=1}^{n}(X_i - \bar{X})^2 \qquad s_Y^2 = \frac{1}{n-1}\sum_{i=1}^{n}(Y_i - \bar{Y})^2$$

Sample Correlation Coefficient

$$R = \frac{[1/(n-1)]\sum_{i=1}^{n}(X_i - \overline{X})(Y_i - \overline{Y})}{\sqrt{[1/(n-1)]\sum_{i=1}^{n}(X_i - \overline{X})^2}\sqrt{[1/(n-1)]\sum_{i=1}^{n}(Y_i - \overline{Y})^2}} = \frac{S_{XY}}{S_X S_Y}.$$

Note that if we compare with the slope estimate $\hat{\beta}_1$ for pairs (X_i, Y_i), we have:

$$\hat{\beta}_1 = \frac{s_{XY}}{s_X^2} = \frac{s_Y}{s_X} R$$

Conditioned on $X_1, \ldots, X_n$ (given $X_1, \ldots, X_n$), under $H_0: \rho = 0 \Rightarrow \beta_1 = 0$:

$$\hat{\beta}_1 = \frac{s_Y}{s_X} R \sim N\left(0, \frac{\sigma_Y^2}{\sum(X_i - \bar{X})^2}\right)$$

Sample Correlation Coefficient

Moreover, we have conditional on $X_i, Var(Y_i|X_i) = (1-\rho^2)\sigma_Y^2$:

$$\frac{\sum_{i=1}^{n}\left(Y_i - \bar{Y} - \hat{\beta}_1(X_i - \bar{X})\right)^2}{(1-\rho^2)\sigma_Y^2} \sim \chi^2(n-2)$$

Since $\hat{\beta}_1 = \frac{s_{XY}}{s_X^2}$ and $s_{XY} = s_Y s_X R$

$$\frac{\sum_{i=1}^{n}\left(Y_i - \bar{Y} - \hat{\beta}_1(X_i - \bar{X})\right)^2}{(1-\rho^2)\sigma_Y^2} = \frac{(n-1)s_Y^2(1-R^2)}{(1-\rho^2)\sigma_Y^2}$$

$\Rightarrow \frac{(n-1)s_Y^2(1-R^2)}{\sigma_Y^2} \sim \chi^2\ (n-2)$ under H_0.

$\hat{\beta}_1 = \frac{s_Y}{s_X}R \sim N\left(0, \frac{\sigma_Y^2}{\sum(X_i-\bar{X})^2}\right)$ under H_0.

$$\Rightarrow \frac{\frac{s_Y}{s_X}R\sqrt{(n-1)s_X^2}}{\sqrt{(n-1)s_Y^2(1-R^2)/n-2}} \sim t(n-2)$$

$$\Rightarrow \frac{R\sqrt{n-2}}{\sqrt{1-R^2}} \sim t(n-2)$$

Sample Correlation Coefficient

Conditioned on $X_1, \ldots, X_n$ (given $X_1, \ldots, X_n$), under $H_0: \rho = 0 \Rightarrow \beta_1 = 0$:

$$\frac{R\sqrt{n-2}}{\sqrt{1-R^2}} \sim t(n-2)$$

Since this conditional distribution does not depend on $X_1, \ldots, X_n$, the unconditional distribution is also $t(n-2)$

Testing correlation

To test $H_0 : \rho = 0$, we have two options:

1. Compute r, the sample correlation coefficient,

$$\boxed{r = \frac{[1/(n-1)] \sum_{i=1}^{n} (x_i - \bar{x})(y_i - \bar{y})}{\sqrt{[1/(n-1)] \sum_{i=1}^{n} (x_i - \bar{x})^2} \sqrt{[1/(n-1)] \sum_{i=1}^{n} (y_i - \bar{y})^2}} = \frac{s_{XY}}{s_X s_Y}},$$

   and then compute

$$\boxed{\frac{r\sqrt{n-2}}{\sqrt{1-r^2}}}.$$

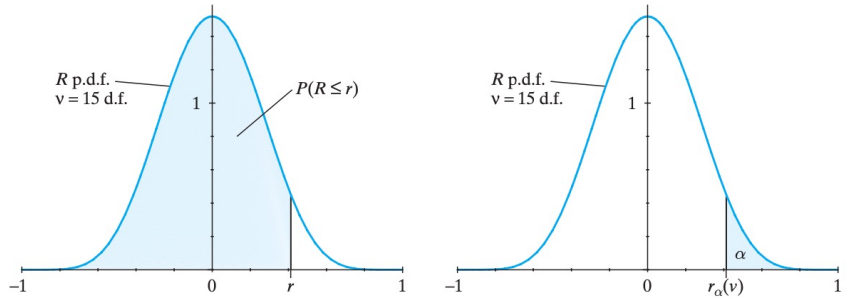
   Compare this value with a $t(n-2)$ quantile using the t-table.

   $t_{\alpha/2}(n-2)$

2. Compute r and use Table IX in Appendix B of the book with $n-2$ degree of freedom.

   $r_{\alpha/2}(n-2)$

Table IX Distribution Function of the Correlation Coefficient R, $\rho = 0$



$$P(R \le r) = \int_{-1}^{r} \frac{\Gamma[(n-1)/2]}{\Gamma(1/2)\Gamma[(n-2)/2]} (1-w^2)^{(n-4)/2}\, dw$$

$\nu = n-2$ degrees of freedom	$P(R \le r)$			
	0.95	0.975	0.99	0.995
	$r_{0.05}(\nu)$	$r_{0.025}(\nu)$	$r_{0.01}(\nu)$	$r_{0.005}(\nu)$
1	0.9877	0.9969	0.9995	0.9999
2	0.9000	0.9500	0.9800	0.9900
3	0.8053	0.8783	0.9343	0.9587
4	0.7292	0.8113	0.8822	0.9172
5	0.6694	0.7544	0.8329	0.8745
6	0.6215	0.7067	0.7887	0.8343
7	0.5822	0.6664	0.7497	0.7977
8	0.5493	0.6319	0.7154	0.7646
9	0.5214	0.6020	0.6850	0.7348
10	0.4972	0.5759	0.6581	0.7079
11	0.4761	0.5529	0.6338	0.6835
12	0.4575	0.5323	0.6120	0.6613
13	0.4408	0.5139	0.5922	0.6411
14	0.4258	0.4973	0.5742	0.6226
15	0.4123	0.4821	0.5577	0.6054
16	0.4000	0.4683	0.5425	0.5897
17	0.3887	0.4555	0.5285	0.5750
18	0.3783	0.4437	0.5154	0.5614
19	0.3687	0.4328	0.5033	0.5487
20	0.3597	0.4226	0.4920	0.5367
25	0.3232	0.3808	0.4450	0.4869
30	0.2959	0.3494	0.4092	0.4487
35	0.2746	0.3246	0.3809	0.4182
40	0.2572	0.3044	0.3578	0.3931
45	0.2428	0.2875	0.3383	0.3721
50	0.2306	0.2732	0.3218	0.3541
60	0.2108	0.2500	0.2948	0.3248
70	0.1954	0.2318	0.2736	0.3017
80	0.1829	0.2172	0.2565	0.2829
90	0.1725	0.2049	0.2422	0.2673
100	0.1638	0.1946	0.2300	0.2540

$$G(r) = P(R \le r) = P\left(T \le \frac{r\sqrt{n-2}}{\sqrt{1-r^2}}\right)$$

$$= \int_{-\infty}^{r\sqrt{n-2}/\sqrt{1-r^2}} h(t)\, dt$$

$$= \int_{-\infty}^{r\sqrt{n-2}/\sqrt{1-r^2}} \frac{\Gamma[(n-1)/2]}{\Gamma(1/2)\,\Gamma[(n-2)/2]} \frac{1}{\sqrt{n-2}} \left(1 + \frac{t^2}{n-2}\right)^{-(n-1)/2} dt.$$

PDF of t(n-2)

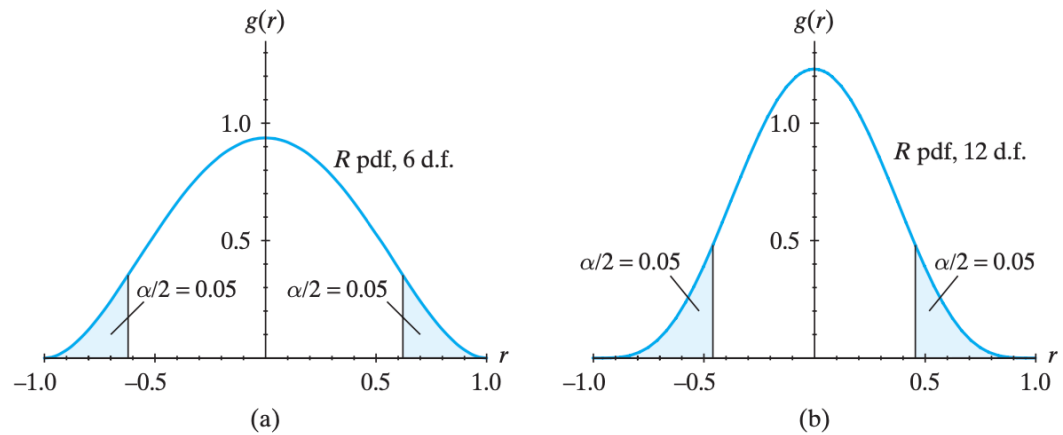


Figure 9.6-1 R pdfs when $n = 8$ and $n = 14$

Summary: testing correlation

Null Hypothesis

$$H_0: \rho = 0$$

Test Statistic

$$t = \frac{R\sqrt{n-2}}{\sqrt{1-R^2}}$$

Distribution under H_0

$$t \sim t(n-2)$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)	p-value
$H_1: \rho \neq 0$	$\lvert t \rvert > t_{\alpha/2}(n-2)$	$2\mathrm{P}(T > \lvert t \rvert)$
$H_1: \rho > 0$	$t > t_{\alpha}(n-2)$	$\mathrm{P}(T > t)$
$H_1: \rho < 0$	$t < -t_{\alpha}(n-2)$	$\mathrm{P}(T < t)$

H_0 is rejected when p-value<α

Test Statistic

$$R$$

Alternative Hypothesis	Rejection/Critical Region (H_0 is rejected)
$H_1: \rho \neq 0$	$\lvert R \rvert > r_{\alpha/2}(n-2)$
$H_1: \rho > 0$	$R > r_{\alpha}(n-2)$
$H_1: \rho < 0$	$R < -r_{\alpha}(n-2)$

Testing correlation

What if we want to test $H_0 : \rho = \rho_0$ for some $\rho_0 \neq 0$? It turns out that we can prove

$$\frac{1}{2}\ln\left(\frac{1+R}{1-R}\right) \overset{approx}{\sim} N\left(\frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right), \frac{1}{n-3}\right).$$

As a result, we can use

$$\boxed{\frac{\frac{1}{2}\ln\left(\frac{1+R}{1-R}\right) - \frac{1}{2}\ln\left(\frac{1+\rho_0}{1-\rho_0}\right)}{\sqrt{1/(n-3)}}}$$

as our test statistic, and compare this with z_α or $-z_\alpha$ or $z_{\alpha/2}$ depending on H_1.

Example

the following pair of data:

y_i	3.1	4.4	4.3	6.1	6.7	7.3
x_i	0	0	0	1	1	1

$\overline{x} = 0.5$

$\overline{y} = 5.317$

$s_X^2 = 0.3$

$s_Y^2 = 2.650$

$$s_{XY} = \frac{1}{6-1}\sum_{i=1}^{6}(y_i - \overline{y})(x_i - \overline{x}) = 0.83$$

$$r = \frac{s_{XY}}{s_X s_Y} = 0.931$$

$$\Rightarrow \quad \frac{r\sqrt{n-2}}{\sqrt{1-r^2}} = 5.099$$

To test $H_0 : \rho = 0$ against $H_1 : \rho \neq 0$ at $\alpha = 0.05$:

1. $$\left|\frac{r\sqrt{n-2}}{\sqrt{1-r^2}}\right| = |5.099| = 5.099 > t_{0.025}(6-2) = t_{0.025}(4) = 2.776$$

   so H_0 is rejected at $\alpha = 5\%$.

2. $r = 0.931$. From Table IX in Appendix B, the critical value is $r_{0.025}(4) = 0.8113$. We reject H_0 since $|r| = 0.931 > 0.8113$.

Example

the following pair of data:

y_i	3.1	4.4	4.3	6.1	6.7	7.3
x_i	0	0	0	1	1	1

To test $H_0 : \rho = 0.5$ against $H_1 : \rho \neq 0.5$ at $\alpha = 0.05$, we compute

$$\frac{\frac{1}{2}\ln\left(\frac{1+0.931}{1-0.931}\right) - \frac{1}{2}\ln\left(\frac{1+0.5}{1-0.5}\right)}{\sqrt{1/(6-3)}} = 1.93$$

Since $|1.93| = 1.93 < 1.96 = z_{0.025}$, we fail to reject H_0 at 5%.

(approximate) Confidence interval for ρ

$$\frac{1}{2}\ln\left(\frac{1+R}{1-R}\right) \overset{approx}{\sim} N\left(\frac{1}{2}\ln\left(\frac{1+\rho}{1-\rho}\right), \frac{1}{n-3}\right)$$

$$P\left(-c \le \frac{(1/2)\ln[(1+R)/(1-R)] - (1/2)\ln[(1+\rho)/(1-\rho)]}{\sqrt{1/(n-3)}} \le c\right) \approx 1-\alpha.$$

$$c = z_{\alpha/2}$$

$$P\left(\frac{1+R-(1-R)\exp(2c/\sqrt{n-3})}{1+R+(1-R)\exp(2c/\sqrt{n-3})} \le \rho \le \frac{1+R-(1-R)\exp(-2c/\sqrt{n-3})}{1+R+(1-R)\exp(-2c/\sqrt{n-3})}\right) \approx 1-\alpha.$$

(approximate) Confidence interval for ρ

Example 9.6-3

Suppose that a random sample of size 12 from a bivariate normal distribution yielded a correlation coefficient of $r = 0.6$. An approximate 95% confidence interval for ρ would be

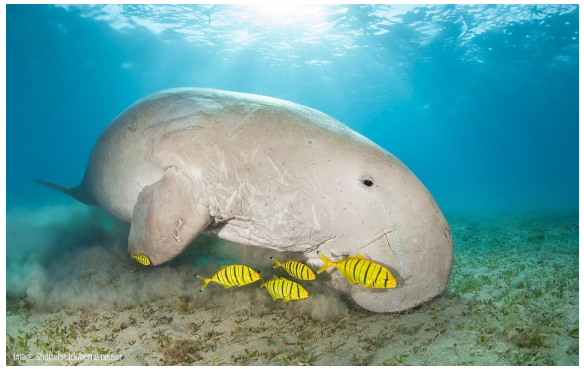
$$\left[\frac{1+0.6-(1-0.6)\exp\left(\frac{2(1.96)}{3}\right)}{1+0.6+(1-0.6)\exp\left(\frac{2(1.96)}{3}\right)}, \frac{1+0.6-(1-0.6)\exp\left(\frac{-2(1.96)}{3}\right)}{1+0.6+(1-0.6)\exp\left(\frac{-2(1.96)}{3}\right)}\right]$$

$$= [0.040, 0.873].$$

If the sample size had been $n = 39$ and $r = 0.6$, the approximate 95% confidence interval would have been $[0.351, 0.770]$. ■

Quiz

11-15. In an article in *Statistics and Computing* ["An Iterative Monte Carlo Method for Nonconjugate Bayesian Analysis" (1991, pp. 119–128)] Carlin and Gelfand investigated the age (x) and length (y) of 27 captured dugongs (sea cows).



$x =$ 1.0, 1.5, 1.5, 1.5, 2.5, 4.0, 5.0, 5.0, 7.0, 8.0, 8.5, 9.0, 9.5, 9.5, 10.0, 12.0, 12.0, 13.0, 13.0, 14.5, 15.5, 15.5, 16.5, 17.0, 22.5, 29.0, 31.5

$y =$ 1.80, 1.85, 1.87, 1.77, 2.02, 2.27, 2.15, 2.26, 2.47, 2.19, 2.26, 2.40, 2.39, 2.41, 2.50, 2.32, 2.32, 2.43, 2.47, 2.56, 2.65, 2.47, 2.64, 2.56, 2.70, 2.72, 2.57

More example

11-6. The following table presents the highway gasoline mileage performance and engine displacement for Daimler-Chrysler vehicles for model year 2005 (source: U.S. Environmental Protection Agency).

(a) Fit a simple linear model relating highway miles per gallon (y) to engine displacement (x) in cubic inches using least squares.

(b) Find an estimate of the mean highway gasoline mileage performance for a car with 150 cubic inches engine displacement.

(c) Obtain the fitted value of y and the corresponding residual for a car, the Neon, with an engine displacement of 122 cubic inches.

Carline	Engine Displacement (in^3)	MPG (highway)
300C/SRT-8	215	30.8
CARAVAN 2WD	201	32.5
CROSSFIRE ROADSTER	196	35.4
DAKOTA PICKUP 2WD	226	28.1
DAKOTA PICKUP 4WD	226	24.4
DURANGO 2WD	348	24.1
GRAND CHEROKEE 2WD	226	28.5
GRAND CHEROKEE 4WD	348	24.2
LIBERTY/CHEROKEE 2WD	148	32.8
LIBERTY/CHEROKEE 4WD	226	28
NEON/SRT-4/SX 2.0	122	41.3
PACIFICA 2WD	215	30.0
PACIFICA AWD	215	28.2
PT CRUISER	148	34.1
RAM 1500 PICKUP 2WD	500	18.7
RAM 1500 PICKUP 4WD	348	20.3
SEBRING 4-DR	165	35.1
STRATUS 4-DR	148	37.9
TOWN & COUNTRY 2WD	148	33.8
VIPER CONVERTIBLE	500	25.9
WRANGLER/TJ 4WD	148	26.4

Not Required

Extension: Multiple Linear Regression

Not Required

Multiple Linear Regression

- Simple linear regression: one variable x
- Multiple linear regression: multiple variables x_1, x_2, …, x_k
- Example:
- simple linear regression

  property tax = a*house price + b
- multiple linear regression

  property tax = a_1*house price + a_2*house size + b

- Question: how to fit multiple linear regression model?

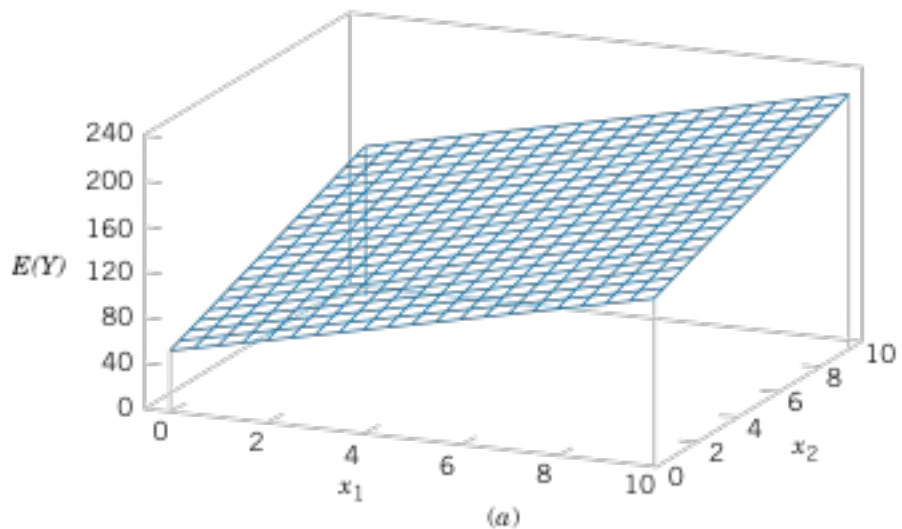
Not Required

- Multiple linear regression model with two variables

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

dependent variable **independent**

response **regressor variables**



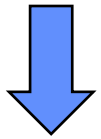
$$E(Y) = 50 + 10x_1 + 7x_2$$

Not Required

Method-of-least-square

- **Estimating coefficiences by least-square**

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \sum_{i=1}^{n} \left(y_i - \beta_0 - \sum_{j=1}^{k} \beta_j x_{ij} \right)^2$$



$$\left. \frac{\partial L}{\partial \beta_0} \right|_{\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k} = -2 \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \sum_{j=1}^{k} \hat{\beta}_j x_{ij} \right) = 0$$

$$\left. \frac{\partial L}{\partial \beta_j} \right|_{\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k} = -2 \sum_{i=1}^{n} \left(y_i - \hat{\beta}_0 - \sum_{j=1}^{k} \hat{\beta}_j x_{ij} \right) x_{ij} = 0 \quad j = 1, 2, \ldots, k$$

Normal equation

Not Required

Least square normal equations

$$
\begin{aligned}
n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_{i1} + \hat{\beta}_2 \sum_{i=1}^{n} x_{i2} + \cdots + \hat{\beta}_k \sum_{i=1}^{n} x_{ik} &= \sum_{i=1}^{n} y_i \\
\hat{\beta}_0 \sum_{i=1}^{n} x_{i1} + \hat{\beta}_1 \sum_{i=1}^{n} x_{i1}^2 + \hat{\beta}_2 \sum_{i=1}^{n} x_{i1} x_{i2} + \cdots + \hat{\beta}_k \sum_{i=1}^{n} x_{i1} x_{ik} &= \sum_{i=1}^{n} x_{i1} y_i \\
\vdots \qquad\qquad \vdots \qquad\qquad \vdots \qquad\qquad \vdots \qquad\qquad \vdots \qquad\qquad \vdots \\
\hat{\beta}_0 \sum_{i=1}^{n} x_{ik} + \hat{\beta}_1 \sum_{i=1}^{n} x_{ik} x_{i1} + \hat{\beta}_2 \sum_{i=1}^{n} x_{ik} x_{i2} + \cdots + \hat{\beta}_k \sum_{i=1}^{n} x_{ik}^2 &= \sum_{i=1}^{n} x_{ik} y_i
\end{aligned}
$$

k+1 equations, k+1 variables.

The solution of the equation will be least-square estimator for the coefficients $\hat{\beta}_0, \hat{\beta}_1, \ldots, \hat{\beta}_k$.

Not Required

Example

EXAMPLE 12-1 Wire Bond Strength

In Chapter 1, we used data on pull strength of a wire bond in a semiconductor manufacturing process, wire length, and die height to illustrate building an empirical model. We will use the same data, repeated for convenience in Table 12-2, and show the details of estimating the model parameters. A three-dimensional scatter plot of the data is presented in Fig. 1-15. Figure 12-4 shows a matrix of two-dimensional scatter plots of the data. These displays can be helpful in visualizing the relationships among variables in a multivariable data set. For example, the plot indicates that there is a strong linear relationship between strength and wire length.

Specifically, we will fit the multiple linear regression model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

where Y = pull strength, x_1 = wire length, and x_2 = die height. From the data in Table 12-2 we calculate

$$n = 25, \sum_{i=1}^{25} y_i = 725.82$$

$$\sum_{i=1}^{25} x_{i1} = 206, \sum_{i=1}^{25} x_{i2} = 8{,}294$$

$$\sum_{i=1}^{25} x_{i1}^2 = 2{,}396, \sum_{i=1}^{25} x_{i2}^2 = 3{,}531{,}848$$

$$\sum_{i=1}^{25} x_{i1}x_{i2} = 77{,}177, \sum_{i=1}^{25} x_{i1}y_i = 8{,}008.47,$$

$$\sum_{i=1}^{25} x_{i2}y_i = 274{,}816.71$$

Not Required

Example: Data

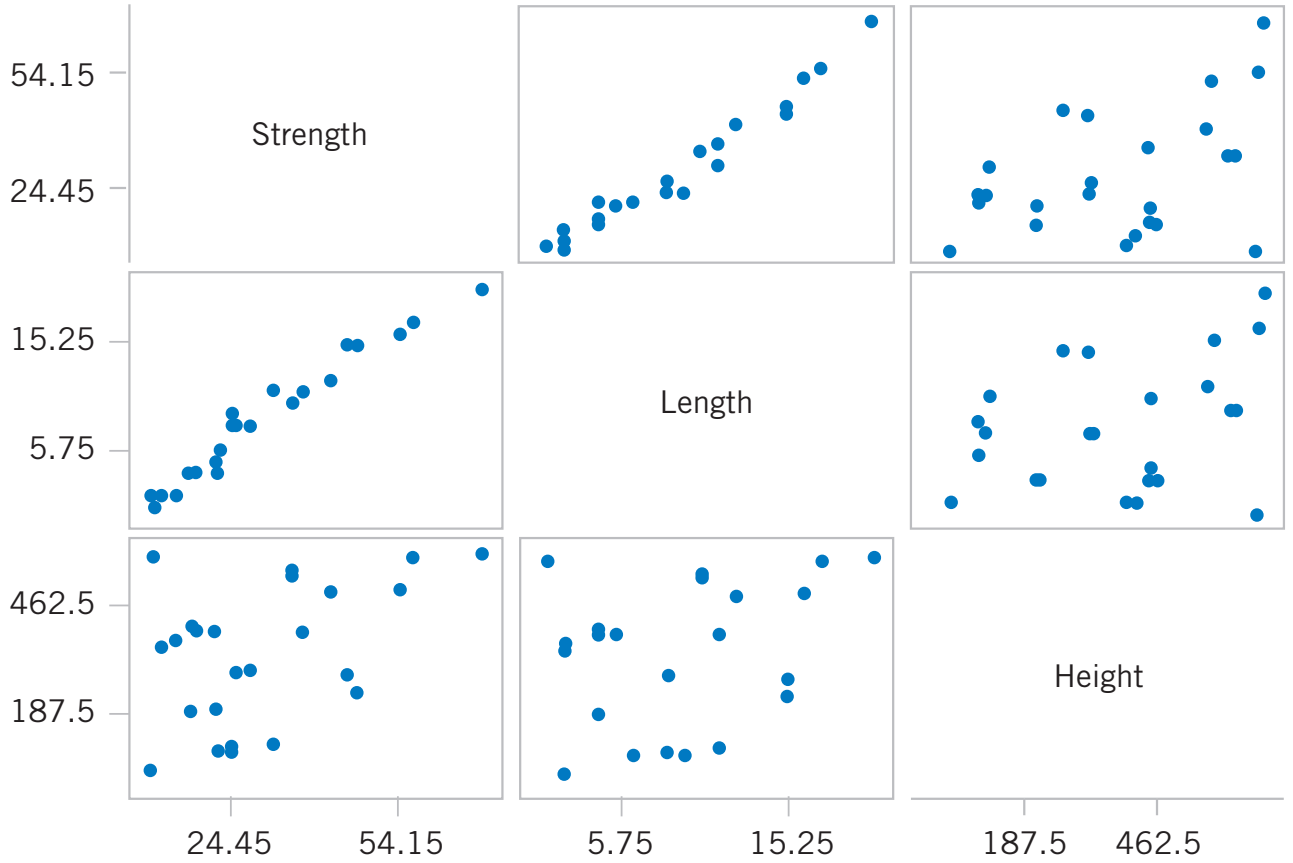
Table 12-2 Wire Bond Data for Example 12-1

Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2	Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2
1	9.95	2	50	14	11.66	2	360
2	24.45	8	110	15	21.65	4	205
3	31.75	11	120	16	17.89	4	400
4	35.00	10	550	17	69.00	20	600
5	25.02	8	295	18	10.30	1	585
6	16.86	4	200	19	34.93	10	540
7	14.38	2	375	20	46.59	15	250
8	9.60	2	52	21	44.88	15	290
9	24.35	9	100	22	54.12	16	510
10	27.50	8	300	23	56.63	17	590
11	17.08	4	412	24	22.13	6	100
12	37.00	11	400	25	21.15	5	400
13	41.95	12	500				

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

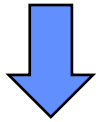
Not Required

Scatter plot



Not Required

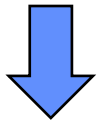
$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$



$$n\hat{\beta}_0 + \hat{\beta}_1 \sum_{i=1}^{n} x_{i1} + \hat{\beta}_2 \sum_{i=1}^{n} x_{i2} = \sum_{i=1}^{n} y_i$$

$$\hat{\beta}_0 \sum_{i=1}^{n} x_{i1} + \hat{\beta}_1 \sum_{i=1}^{n} x_{i1}^2 + \hat{\beta}_2 \sum_{i=1}^{n} x_{i1}x_{i2} = \sum_{i=1}^{n} x_{i1}y_i$$

$$\hat{\beta}_0 \sum_{i=1}^{n} x_{i2} + \hat{\beta}_1 \sum_{i=1}^{n} x_{i1}x_{i2} + \hat{\beta}_2 \sum_{i=1}^{n} x_{i2}^2 = \sum_{i=1}^{n} x_{i2}y_i$$



$$25\hat{\beta}_0 + 206\hat{\beta}_1 + 8294\hat{\beta}_2 = 725.82$$

$$206\hat{\beta}_0 + 2396\hat{\beta}_1 + 77{,}177\hat{\beta}_2 = 8{,}008.47$$

$$8294\hat{\beta}_0 + 77{,}177\hat{\beta}_1 + 3{,}531{,}848\hat{\beta}_2 = 274{,}816.71$$

Not Required

- **Solution**

$$\hat{\beta}_0 = 2.26379, \quad \hat{\beta}_1 = 2.74427, \quad \hat{\beta}_2 = 0.01253$$

- **Fitted regression model**

$$\hat{y} = 2.26379 + 2.74427x_1 + 0.01253x_2$$

$$\widehat{\text{Pull strength}} = 2.26 + 2.74(\text{wire length}) + 0.0125(\text{die height})$$

Data

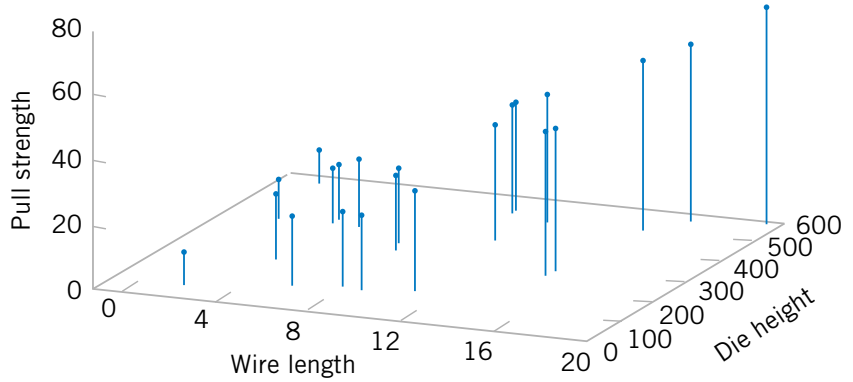


Figure 1-15 Three-dimensional plot of the wire bond pull strength data.

Model

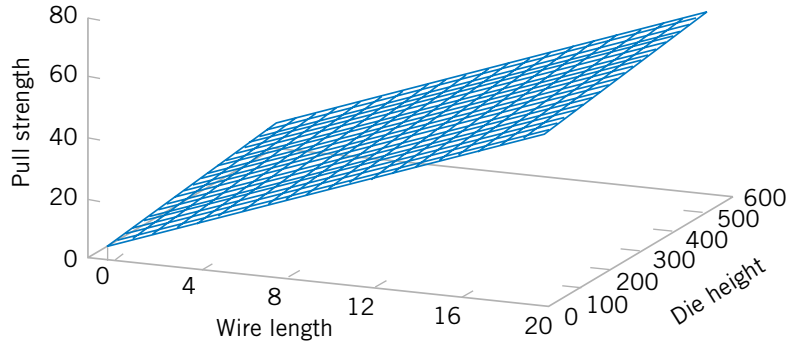


Figure 1-16 Plot of predicted values of pull strength from the empirical model.

Not Required

Matrix Approach

$$y_i = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_k x_{ik} + \epsilon_i \qquad i = 1, 2, \ldots, n$$

$$\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\epsilon} \qquad p = k + 1$$

$$\mathbf{y} = \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} 1 & x_{11} & x_{12} & \cdots & x_{1k} \\ 1 & x_{21} & x_{22} & \cdots & x_{2k} \\ \vdots & \vdots & \vdots & & \vdots \\ 1 & x_{n1} & x_{n2} & \cdots & x_{nk} \end{bmatrix} \quad \boldsymbol{\beta} = \begin{bmatrix} \beta_0 \\ \beta_1 \\ \vdots \\ \beta_k \end{bmatrix} \quad \text{and} \quad \boldsymbol{\epsilon} = \begin{bmatrix} \epsilon_1 \\ \epsilon_2 \\ \vdots \\ \epsilon_n \end{bmatrix}$$

$(n \times 1)$ $(n \times p)$ $(p \times 1)$ $(n \times 1)$

Goal: Solve for β given X and y

Not Required

Solution

$$L = \sum_{i=1}^{n} \epsilon_i^2 = \boldsymbol{\epsilon}'\boldsymbol{\epsilon} = (\mathbf{y} - \mathbf{X}\boldsymbol{\beta})'(\mathbf{y} - \mathbf{X}\boldsymbol{\beta})$$

$$\frac{\partial L}{\partial \boldsymbol{\beta}} = \mathbf{0} \Rightarrow$$

Normal Equations

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$$

Least Square Estimate of β

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

$$\begin{bmatrix} n & \sum_{i=1}^{n} x_{i1} & \sum_{i=1}^{n} x_{i2} & \cdots & \sum_{i=1}^{n} x_{ik} \\ \sum_{i=1}^{n} x_{i1} & \sum_{i=1}^{n} x_{i1}^2 & \sum_{i=1}^{n} x_{i1}x_{i2} & \cdots & \sum_{i=1}^{n} x_{i1}x_{ik} \\ \vdots & \vdots & \vdots & & \vdots \\ \sum_{i=1}^{n} x_{ik} & \sum_{i=1}^{n} x_{ik}x_{i1} & \sum_{i=1}^{n} x_{ik}x_{i2} & \cdots & \sum_{i=1}^{n} x_{ik}^2 \end{bmatrix} \begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \vdots \\ \hat{\beta}_k \end{bmatrix} = \begin{bmatrix} \sum_{i=1}^{n} y_i \\ \sum_{i=1}^{n} x_{i1}y_i \\ \vdots \\ \sum_{i=1}^{n} x_{ik}y_i \end{bmatrix}$$

Not Required

Simple example

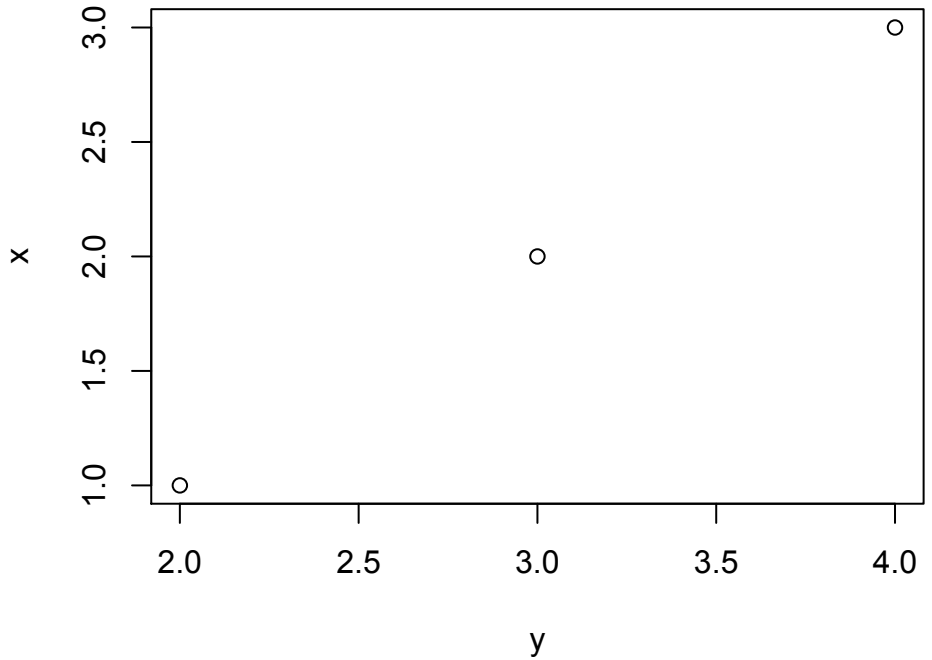
- One variable
- 3 data points

(1, 2), (2, 3), (3, 4)

- Write down the normal equation and solve for the coefficients

$$\begin{bmatrix} 3 & 6 \\ 6 & 14 \end{bmatrix} \begin{bmatrix} \beta_0 \\ \beta_1 \end{bmatrix} = \begin{bmatrix} 9 \\ 20 \end{bmatrix}$$

$$\beta_0 = 1, \beta_1 = 1$$



Not Required

Example: Wire Bond Data

Table 12-2 Wire Bond Data for Example 12-1

Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2	Observation Number	Pull Strength y	Wire Length x_1	Die Height x_2
1	9.95	2	50	14	11.66	2	360
2	24.45	8	110	15	21.65	4	205
3	31.75	11	120	16	17.89	4	400
4	35.00	10	550	17	69.00	20	600
5	25.02	8	295	18	10.30	1	585
6	16.86	4	200	19	34.93	10	540
7	14.38	2	375	20	46.59	15	250
8	9.60	2	52	21	44.88	15	290
9	24.35	9	100	22	54.12	16	510
10	27.50	8	300	23	56.63	17	590
11	17.08	4	412	24	22.13	6	100
12	37.00	11	400	25	21.15	5	400
13	41.95	12	500				

Not Required

Example

$$\mathbf{X} = \begin{bmatrix} 1 & 2 & 50 \\ 1 & 8 & 110 \\ 1 & 11 & 120 \\ 1 & 10 & 550 \\ 1 & 8 & 295 \\ 1 & 4 & 200 \\ 1 & 2 & 375 \\ 1 & 2 & 52 \\ 1 & 9 & 100 \\ 1 & 8 & 300 \\ 1 & 4 & 412 \\ 1 & 11 & 400 \\ 1 & 12 & 500 \\ 1 & 2 & 360 \\ 1 & 4 & 205 \\ 1 & 4 & 400 \\ 1 & 20 & 600 \\ 1 & 1 & 585 \\ 1 & 10 & 540 \\ 1 & 15 & 250 \\ 1 & 15 & 290 \\ 1 & 16 & 510 \\ 1 & 17 & 590 \\ 1 & 6 & 100 \\ 1 & 5 & 400 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 9.95 \\ 24.45 \\ 31.75 \\ 35.00 \\ 25.02 \\ 16.86 \\ 14.38 \\ 9.60 \\ 24.35 \\ 27.50 \\ 17.08 \\ 37.00 \\ 41.95 \\ 11.66 \\ 21.65 \\ 17.89 \\ 69.00 \\ 10.30 \\ 34.93 \\ 46.59 \\ 44.88 \\ 54.12 \\ 56.63 \\ 22.13 \\ 21.15 \end{bmatrix}$$

$$\mathbf{X}'\mathbf{X}\hat{\boldsymbol{\beta}} = \mathbf{X}'\mathbf{y}$$

$$\mathbf{X}'\mathbf{X} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 2 & 8 & \cdots & 5 \\ 50 & 110 & \cdots & 400 \end{bmatrix} \begin{bmatrix} 1 & 2 & 50 \\ 1 & 8 & 110 \\ \vdots & \vdots & \vdots \\ 1 & 5 & 400 \end{bmatrix}$$

$$= \begin{bmatrix} 25 & 206 & 8{,}294 \\ 206 & 2{,}396 & 77{,}177 \\ 8{,}294 & 77{,}177 & 3{,}531{,}848 \end{bmatrix}$$

$$\mathbf{X}'\mathbf{y} = \begin{bmatrix} 1 & 1 & \cdots & 1 \\ 2 & 8 & \cdots & 5 \\ 50 & 110 & \cdots & 400 \end{bmatrix} \begin{bmatrix} 9.95 \\ 24.45 \\ \vdots \\ 21.15 \end{bmatrix} = \begin{bmatrix} 725.82 \\ 8{,}008.47 \\ 274{,}816.71 \end{bmatrix}$$

Not Required

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}'\mathbf{X})^{-1}\mathbf{X}'\mathbf{y}$$

$$\begin{bmatrix} \hat{\beta}_0 \\ \hat{\beta}_1 \\ \hat{\beta}_2 \end{bmatrix} = \begin{bmatrix} 25 & 206 & 8{,}294 \\ 206 & 2{,}396 & 77{,}177 \\ 8{,}294 & 77{,}177 & 3{,}531{,}848 \end{bmatrix}^{-1} \begin{bmatrix} 725.82 \\ 8{,}008.37 \\ 274{,}811.31 \end{bmatrix}$$

$$= \begin{bmatrix} 0.214653 & -0.007491 & -0.000340 \\ -0.007491 & 0.001671 & -0.000019 \\ -0.000340 & -0.000019 & +0.0000015 \end{bmatrix} \begin{bmatrix} 725.82 \\ 8{,}008.47 \\ 274{,}811.31 \end{bmatrix}$$

$$= \begin{bmatrix} 2.26379143 \\ 2.74426964 \\ 0.01252781 \end{bmatrix}$$

$$\hat{y} = 2.26379 + 2.74427x_1 + 0.01253x_2$$

Not Required

$$x_{11} = 2 \quad x_{12} = 50$$

$$\begin{aligned}\hat{y}_1 &= 2.26379 + 2.74427x_{11} + 0.01253x_{12} \\ &= 2.26379 + 2.74427(2) + 0.01253(50) \\ &= 8.38\end{aligned}$$

The corresponding observed value is $y_1 = 9.95$. The *residual* corresponding to the first observation is

$$\begin{aligned}e_1 &= y_1 - \hat{y}_1 \\ &= 9.95 - 8.38 \\ &= 1.57\end{aligned}$$

Not Required

Example

12-4. You have fit a multiple linear regression model and the $(\mathbf{X}'\mathbf{X})^{-1}$ matrix is:

$$(\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} 0.893758 & -0.0282448 & -0.0175641 \\ -0.028245 & 0.0013329 & 0.0001547 \\ -0.017564 & 0.0001547 & 0.0009108 \end{bmatrix}$$

(a) How many regressor variables are in this model?

(b) If the error sum of squares is 307 and there are 15 observations, what is the estimate of σ^2?

(c) What is the standard error of the regression coefficient $\hat{\beta}_1$?

Not Required

Residual analysis

- **Estimate the variance**

Estimating σ^2

$$\hat{\sigma}^2 = \frac{\sum_{i=1}^{n} e_i^2}{n-p} = \frac{SS_E}{n-p}$$

$$\mathbf{e} = \mathbf{y} - \hat{\mathbf{y}} = \mathbf{y} - \mathbf{X}\hat{\boldsymbol{\beta}}$$

Not Required

(cont.) wire-bound example

Wire example $\hat{\sigma}^2 = 115.2/22 = 5.2364.$

Table 12-3 Observations, Fitted Values, and Residuals for Example 12-2

Observation Number	y_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$	Observation Number	y_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$
1	9.95	8.38	1.57	14	11.66	12.26	−0.60
2	24.45	25.60	−1.15	15	21.65	15.81	5.84
3	31.75	33.95	−2.20	16	17.89	18.25	−0.36
4	35.00	36.60	−1.60	17	69.00	64.67	4.33
5	25.02	27.91	−2.89	18	10.30	12.34	−2.04
6	16.86	15.75	1.11	19	34.93	36.47	−1.54
7	14.38	12.45	1.93	20	46.59	46.56	0.03
8	9.60	8.40	1.20	21	44.88	47.06	−2.18
9	24.35	28.21	−3.86	22	54.12	52.56	1.56
10	27.50	27.98	−0.48	23	56.63	56.31	0.32
11	17.08	18.40	−1.32	24	22.13	19.98	2.15
12	37.00	37.46	−0.46	25	21.15	21.00	0.15
13	41.95	41.46	0.49				

Not Required

Covariance matrix

- **Unbiased estimator** $E(\hat{\boldsymbol{\beta}}) = \boldsymbol{\beta}$
- **Covariance matrix of regression coefficient**

$$\text{cov}(\hat{\boldsymbol{\beta}}) = \sigma^2(\mathbf{X}'\mathbf{X})^{-1} = \sigma^2\mathbf{C}$$

estimated standard error of $\hat{\beta}_j$ $\quad se(\hat{\beta}_j) = \sqrt{\hat{\sigma}^2 C_{jj}}$

- **e.g., k = 2**

$$\mathbf{C} = (\mathbf{X}'\mathbf{X})^{-1} = \begin{bmatrix} C_{00} & C_{01} & C_{02} \\ C_{10} & C_{11} & C_{12} \\ C_{20} & C_{21} & C_{22} \end{bmatrix}$$

$$V(\hat{\beta}_j) = \sigma^2 C_{jj}, \qquad j = 0, 1, 2$$

$$\text{cov}(\hat{\beta}_i, \hat{\beta}_j) = \sigma^2 C_{ij}, \qquad i \neq j$$

Not Required

Test for individual coefficients

- **Partial (or marginal) test**

$$H_0: \beta_j = \beta_{j0}$$
$$H_1: \beta_j \neq \beta_{j0}$$

$$T_0 = \frac{\hat{\beta}_j - \beta_{j0}}{\sqrt{\sigma^2 C_{jj}}} = \frac{\hat{\beta}_j - \beta_{j0}}{se(\hat{\beta}_j)}$$

- **Test for significance of each coefficient: testing whether a coefficient is zero; if the null is not rejected, then that variable can be deleted from model**

Not Required

Example

EXAMPLE 12-4 **Wire Bond Strength Coefficient Test**

Consider the wire bond pull strength data, and suppose that we want to test the hypothesis that the regression coefficient for x_2 (die height) is zero. The hypotheses are

$$H_0: \beta_2 = 0$$
$$H_1: \beta_2 \neq 0$$

The main diagonal element of the $(\mathbf{X}'\mathbf{X})^{-1}$ matrix corresponding to $\hat{\beta}_2$ is $C_{22} = 0.0000015$, so the t-statistic in Equation 12-25 is

$$t_0 = \frac{\hat{\beta}_2}{\sqrt{\hat{\sigma}^2 C_{22}}} = \frac{0.01253}{\sqrt{(5.2352)(0.0000015)}} = 4.477$$

Table 12-3 Observations, Fitted Values, and Residuals for Example 12-2

Observation Number	y_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$	Observation Number	y_i	$\hat{y}_i$	$e_i = y_i - \hat{y}_i$
1	9.95	8.38	1.57	14	11.66	12.26	−0.60
2	24.45	25.60	−1.15	15	21.65	15.81	5.84
3	31.75	33.95	−2.20	16	17.89	18.25	−0.36
4	35.00	36.60	−1.60	17	69.00	64.67	4.33
5	25.02	27.91	−2.89	18	10.30	12.34	−2.04
6	16.86	15.75	1.11	19	34.93	36.47	−1.54
7	14.38	12.45	1.93	20	46.59	46.56	0.03
8	9.60	8.40	1.20	21	44.88	47.06	−2.18
9	24.35	28.21	−3.86	22	54.12	52.56	1.56
10	27.50	27.98	−0.48	23	56.63	56.31	0.32
11	17.08	18.40	−1.32	24	22.13	19.98	2.15
12	37.00	37.46	−0.46	25	21.15	21.00	0.15
13	41.95	41.46	0.49				

. Since $t_{0.025,22} = 2.074$, we reject $H_0: \beta_2 = 0$

P-value $P = 0.0002$,

Not Required

Confidence intervals

Confidence Interval on a Regression Coefficient

A $100(1-\alpha)\%$ **confidence interval on the regression coefficient** β_j, $j = 0, 1, \ldots, k$ in the multiple linear regression model is given by

$$\hat{\beta}_j - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 C_{jj}} \le \beta_j \le \hat{\beta}_j + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 C_{jj}}$$

Confidence Interval on the Mean Response

For the multiple linear regression model, a $100(1-\alpha)\%$ **confidence interval on the mean response** at the point $x_{01}, x_{02}, \ldots, x_{0k}$ is

$$\hat{\mu}_{Y|\mathbf{x}_0} - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0} \le \mu_{Y|\mathbf{x}_0} \le \hat{\mu}_{Y|\mathbf{x}_0} + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2 \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0}$$

Prediction Interval

A $100(1-\alpha)\%$ **prediction interval for this future observation** is

$$\hat{y}_0 - t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)} \le Y_0 \le \hat{y}_0 + t_{\alpha/2,n-p}\sqrt{\hat{\sigma}^2(1 + \mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0)}$$

Not Required

Example

EXAMPLE 12-8 Wire Bond Strength Confidence Interval on the Mean Response

The engineer in Example 12-1 would like to construct a 95% CI on the mean pull strength for a wire bond with wire length $x_1 = 8$ and die height $x_2 = 275$. Therefore,

$$\mathbf{x}_0 = \begin{bmatrix} 1 \\ 8 \\ 275 \end{bmatrix}$$

The estimated mean response at this point is found from Equation 12-36 as

$$\hat{\mu}_{Y|\mathbf{x}_0} = \mathbf{x}_0'\hat{\boldsymbol{\beta}} = \begin{bmatrix} 1 & 8 & 275 \end{bmatrix} \begin{bmatrix} 2.26379 \\ 2.74427 \\ 0.01253 \end{bmatrix} = 27.66$$

The variance of $\hat{\mu}_{Y|\mathbf{x}_0}$ is estimated by

$$\hat{\sigma}^2\mathbf{x}_0'(\mathbf{X}'\mathbf{X})^{-1}\mathbf{x}_0 = 5.2352 \begin{bmatrix} 1 & 8 & 275 \end{bmatrix} \times \begin{bmatrix} .214653 & -.007491 & -.000340 \\ -.007491 & .001671 & -.000019 \\ -.000340 & -.000019 & .0000015 \end{bmatrix} \begin{bmatrix} 1 \\ 8 \\ 275 \end{bmatrix}$$

$$= 5.2352\,(0.0444) = 0.23244$$

Therefore, a 95% CI on the mean pull strength at this point is found from Equation 12-39 as

$$27.66 - 2.074\sqrt{0.23244} \le \mu_{Y|\mathbf{x}_0} \le 27.66 + 2.074\sqrt{0.23244}$$

which reduces to

$$26.66 \le \mu_{Y|\mathbf{x}_0} \le 28.66$$

Some computer software packages will provide estimates of the mean for a point of interest $\mathbf{x}_0$ and the associated CI. Table 12-4 shows the Minitab output for Example 12-8. Both the estimate of the mean and the 95% CI are provided.

Not Required

OTHER EXTENSIONS

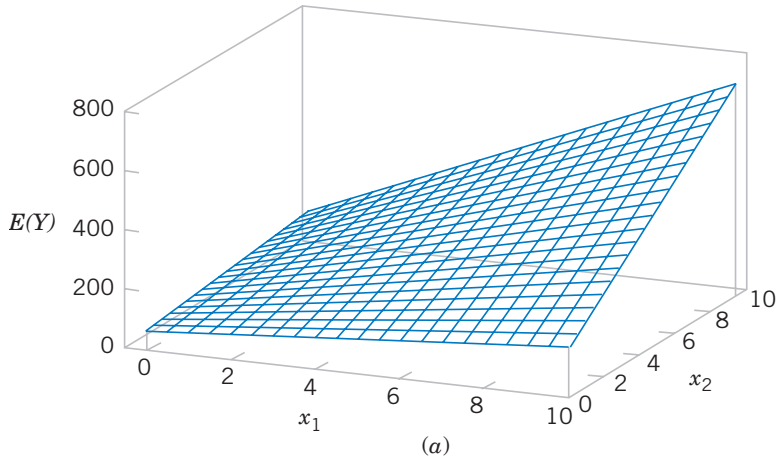
Not Required

Generalization: Linear regression

- **Cubic polynomial**

$$Y = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \epsilon$$

$$\text{let } x_1 = x, x_2 = x^2, x_3 = x^3,$$



$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \beta_3 x_3 + \epsilon$$

- **Interaction between variables**

$$Y = 50 + 10x_1 + 7x_2 + 5x_1 x_2$$

Not Required

Deal with categorical variables

- **Quantitative variable: temperature, pressure, distance, voltage**
- **Categorical variables (qualitative): location, description, type, gender**

$$x = \begin{cases} 0 \text{ if the observation is from operator 1} \\ 1 \text{ if the observation is from operator 2} \end{cases}$$

Indicator variable

Not Required

“One-hot” keying

- **In general, a qualitative variable with r-levels can be modeled by r-1 indicator variable**
- **E.g. for a 3-evel observation, using 2 indicator variables**

x_1	x_2	
0	0	if the observation is from operator 1
1	0	if the observation is from operator 2
0	1	if the observation is from operator 3

Not Required

Example: Categorical

EXAMPLE 12-13 Surface Finish

A mechanical engineer is investigating the surface finish of metal parts produced on a lathe and its relationship to the speed (in revolutions per minute) of the lathe. The data are shown in Table 12-15. Note that the data have been collected using two different types of cutting tools. Since the type of cutting tool likely affects the surface finish, we will fit the model

$$Y = \beta_0 + \beta_1 x_1 + \beta_2 x_2 + \epsilon$$

where Y is the surface finish, x_1 is the lathe speed in revolutions per minute, and x_2 is an indicator variable denoting the type of cutting tool used; that is,

$$x_2 = \begin{cases} 0, \text{ for tool type 302} \\ 1, \text{ for tool type 416} \end{cases}$$



Table 12-15 Surface Finish Data for Example 12-13

Observation Number, i	Surface Finish y_i	RPM	Type of Cutting Tool	Observation Number, i	Surface Finish y_i	RPM	Type of Cutting Tool
1	45.44	225	302	11	33.50	224	416
2	42.03	200	302	12	31.23	212	416
3	50.10	250	302	13	37.52	248	416
4	48.75	245	302	14	37.13	260	416
5	47.92	235	302	15	34.70	243	416
6	47.79	237	302	16	33.92	238	416
7	52.26	265	302	17	32.13	224	416
8	50.52	259	302	18	35.47	251	416
9	45.58	221	302	19	33.49	232	416
10	44.78	218	302	20	32.29	216	416

Not Required

$$\mathbf{X} = \begin{bmatrix} 1 & 225 & 0 \\ 1 & 200 & 0 \\ 1 & 250 & 0 \\ 1 & 245 & 0 \\ 1 & 235 & 0 \\ 1 & 237 & 0 \\ 1 & 265 & 0 \\ 1 & 259 & 0 \\ 1 & 221 & 0 \\ 1 & 218 & 0 \\ 1 & 224 & 1 \\ 1 & 212 & 1 \\ 1 & 248 & 1 \\ 1 & 260 & 1 \\ 1 & 243 & 1 \\ 1 & 238 & 1 \\ 1 & 224 & 1 \\ 1 & 251 & 1 \\ 1 & 232 & 1 \\ 1 & 216 & 1 \end{bmatrix} \quad \mathbf{y} = \begin{bmatrix} 45.44 \\ 42.03 \\ 50.10 \\ 48.75 \\ 47.92 \\ 47.79 \\ 52.26 \\ 50.52 \\ 45.58 \\ 44.78 \\ 33.50 \\ 31.23 \\ 37.52 \\ 37.13 \\ 34.70 \\ 33.92 \\ 32.13 \\ 35.47 \\ 33.49 \\ 32.29 \end{bmatrix}$$

The fitted model is

$$\hat{y} = 14.27620 + 0.14115x_1 - 13.28020x_2$$